University of Western Australia

A Quantum Solution to Graph Similarity Problems

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This thesis is presented for the degree of
Masters of Physics (Quantum Technology and Computing)

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27 November 2025

Acknowledgments

This work was supported by resources provided by the Pawsey Supercomputing Research Centre with funding from the Australian Government and the Government of Western Australia.

I would like to thank my supervisor, Jingbo Wang. Your support throughout the course of my studies has made this journey immensely rewarding. Thank you for your excellent management of the QUISA research group and the quantum optimisation subgroup. By taking on the role of connecting people in the pursuit of knowledge, you formed a community of collaborators which extends well beyond the usual barrier of academics and researchers, bringing quantum computing into the practical world.

I also extend my thanks to Tavis, Edric, Pascal, Aidan, Andrew, and Leo for their support on the project. Tavis for sharing his valuable research, which made this whole project possible. Edric for his many insights into using QuOp and assistance bringing this thesis to a scale I couldn't achieve alone. Pascal for the top-down view of the project as a quantum-HPC hybrid expert. Aiden, Andrew, and Leo, for their collaborative efforts which guided the analysis and generation of these results.

Thank you to Benjamin Rossdeutscher. Our academic journeys have been in parallel for many years, and your personal and professional support have been a key part of not just this thesis but the path here.

Finally, I thank my family and friends for their continuous support throughout my degree. Without you, none of this would have been possible.

Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of about 60 pages inclusive of tables and figures but exclusive of references and appendices.

I authorise the Head of the Department of Physics to make a copy of this thesis to any person judged to have an acceptable reason for access to the information, i.e, for research, study or instruction.

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Signature

Date

Supervisor: Professor Jingbo Wang

Signature

Date

Summary of Student Achievement

This thesis took a rather straightforward path from beginning to end. At the beginning of my degree, Dr. Tavis Bennett presented his novel algorithm, the non-variational quantum walk-based optimisation algorithm. Similar to a few others, I was amazed by the results that he was able to achieve using the NVQWOA, and requested to help benchmark the algorithm. I had done some prior study into graph theory and was strongly motivated by the well-known hardness of graph isomorphism, so I requested to work on graph similarity.

To start the project, I began by building foundational knowledge of the quantum algorithms that the NVQWOA is based upon, the QAOA and QWOA.

When I moved to simulating the NVQWOA, Tavis provided some initial code that accelerated the beginning of the project. From this, I developed an easy-to-use NVQWOA simulation which was highly interpretable and showed early strong results. This simulation allowed me to test individual pairs of graphs on the NVQWOA, but it was constrained by the hardware requirements from running more problems at a larger problem size. At the time, it was unclear whether the scope of the project would grow to require high-performance computing, but in 2025 it became obvious that the current computational hardware was not enough to move the project forward. The QUISA research group has a strong collaboration with the Pawsey Supercomputing Centre, so I began preparing to run the simulation on their Setonix system.

Adapting a highly serialised program into an efficiently parallel program is challenging, but I was lucky enough to receive assistance from Dr. Edric Matwiejew, who guided me in using his QVA simulation package QuOp. The original NVQWOA simulation was completely reconstructed into a parallelised program over several stages. I attempted to push the problem sizes even higher, but computational limits and the super-exponential increase in problem size prevented the project from expanding to much larger problems in time. Instead of narrowing down on simulating bigger problems, I widened my view to investigate the behaviour of the algorithm across many different problem instances, how hyperparameter optimisation could affect the results, and how the problem structure fits with the theoretical conditions for the mixer.

Initial work from this investigation was presented at the Quantum Meets Logistics workshop 2025.

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Abstract

Quantum algorithms offer the potential to address combinatorial optimisation problems that are intractable for classical computers. Among these, the graph similarity problem remains a fundamental challenge with applications across network science, logistics, and bioinformatics. This thesis investigates the performance of the non-variational quantum walk-based optimisation algorithm (NVQWOA) on this problem.

Through state-vector simulations, this work characterises the behaviour of the NVQWOA for multiple problem instances, sizes, and algorithm iterations. The algorithm consistently amplifies the probability of optimal and near-optimal solutions relative to the uniform initial state and outperforms Grover's search algorithm on an equivalent number of iterations, demonstrating that constructive interference concentrates amplitude in regions of high solution quality. Analyses based on Hamming and subshell distance reveal that amplification strength depends on structural proximity to the optimum, reflecting the locality of the quantum walk on the permutation graph. Techniques for numerically optimising the hyperparameters are investigated. Additional investigation of quality gaps and variance trends shows that the solution space structure enhances the effectiveness of amplitude redistribution and satisfies key conditions for efficient optimisation.

These results establish NVQWOA as a promising approach to solving the graph similarity problem. Its sensitivity to solution structure and minimal optimisation requirements suggest potential advantages for near-term quantum devices.

Contents

Chapter 1

Introduction

Physics, as a discipline, is fundamentally concerned with describing and understanding the unknown. This pursuit often takes place at one of three broad frontiers. The first is the frontier of the very large, encompassing astrophysics and cosmology, where the fundamental scales of size, energy, and time far exceed human experience. Here, telescopes and detectors search space for signals that test and refine our models of the universe. The second is the frontier of the very small, the domain of quantum fields and fundamental particles, where the scales involved are many orders of magnitude below those of everyday life. Here, particle accelerators, resonant cavities, and precision experiments probe the limits of matter and interactions.

In recent decades, a third frontier has become increasingly prominent: the intractable. This refers to physical and computational systems whose complexity, rather than their size, places them beyond straightforward description. Such systems are not inaccessible because they are too large or too small, but because the number of interacting components grows combinatorially, producing an overwhelming configuration space. Many of the most pressing challenges in modern science and technology lie on this frontier — from simulating strongly correlated quantum systems to optimising large-scale networks and solving combinatorial search problems. Classical computation struggles with such tasks because the number of possible configurations grows exponentially with system size, and even the most powerful supercomputers cannot exhaustively explore these spaces.

One domain where quantum algorithms may offer an advantage is graph similarity: determining how similar two graphs are, particularly when no labels or correspondences are known between their vertices [3]. Graph similarity is an important aspect of many fields, including network analysis, security, logistics, bioinformatics, and chemical informatics [4, 5, 6, 7]. Similarity measures are a key metric in network analysis, allowing structural changes in communications, logistics, or sensor networks to be detected. Across these domains, the ability to compute graph similarity efficiently could directly enhance research, security, and decision-making capabilities.

However, this is a deeply challenging problem. Calculating similarity between unlabelled graphs is NP-hard: the most common exact measure, graph edit distance, scales super-exponentially with graph size. Even for graphs with as few as 16 nodes, computing the exact distance can take more than 10^3 seconds on standard hardware, and many real-world networks, such as those representing Wikipedia's hyperlink structure, contain millions of nodes [8, 9]. This complexity explosion severely limits the applicability of classical algorithms and prevents meaningful exploration of large or structurally complex graphs. Overcoming this computational barrier requires new approaches.

Quantum computing offers a possible approach. By exploiting superposition and interference, quantum computers can explore exponentially large spaces in ways that classical computers cannot, offering the potential for dramatic speed-ups in certain problem classes. Quantum algorithms have already shown theoretical advantages in unstructured search [10], integer factorisation [11], and simulation, and attention is increasingly turning toward their application in combinatorial optimisation. Among the most prominent approaches is the Quantum Approximate Optimisation Algorithm (QAOA) [12, 13, 14], a variational hybrid algorithm that alternates between cost and mixing unitaries to bias a quantum state toward optimal solutions. QAOA and other variational quantum algorithms (VQAs) are conceptually appealing, but their practical performance depends on the optimisation of large parameter spaces, which introduces substantial classical overhead. The quality of solutions is also highly sensitive to initial parameter choices, and optimisation can require thousands of function evaluations even for shallow circuits [15].

These challenges are compounded when applying QAOA to graph similarity. A straightforward bitstring encoding of all possible vertex mappings between two simple n-vertex graphs requires n^2 qubits, producing a total solution space of size 2^{n^2}. However, feasible solutions consisting of bijective vertex permutations, only consist of a space of size $n!$, meaning that almost all encoded states are invalid. Penalty terms can suppress these infeasible states, but they complicate the cost landscape and introduce additional parameters, exacerbating the classical optimisation burden. Prior work by Pritchard et al. demonstrated the feasibility of QAOA for graph similarity and introduced a compact encoding scheme, but performance improvements required global parameter optimisation over thousands of evaluations and remained limited to graphs with $n \leq 9$ [1]. These results highlight the need for alternative quantum optimisation strategies.

One alternative is the non-variational quantum walk-based optimisation algorithm (NVQWOA) [16, 2]. NVQWOA dispenses with the variational loop entirely, instead evolving the system under predetermined unitaries consisting of a quality-based phase separator and a problem-specific mixer. The latter is particularly significant: unlike the transverse-field mixer used in QAOA, which flips individual qubits without regard for problem constraints, using a permutation-based mixer generates transitions directly between feasible solutions. This reduces the accessible state space from 2^{n^2} to $n!$, embedding the combinatorial structure of the problem into the dynamics of the quantum walk itself. NVQWOA depends on only three hyperparameters, γ, t, and β, which are interpretable and show relatively smooth scaling behaviour. This simplicity greatly reduces the classical overhead of parameter selection, while the continuous-time evolution retains the key advantage of quantum interference: amplitude is redistributed from the uniform initial state into regions of the solution space that correspond to high-quality solutions.

Beyond its standalone potential, NVQWOA may also complement classical and machine learning approaches. Graph neural networks (GNNs) have achieved remarkable success in structural learning tasks such as node classification, link prediction, and graph-level regression[17]. However, GNNs have demanding training and data requirements, are domain-specific, and their performance often degrades when applied to unlabelled or permuted graphs, which is precisely where graph similarity is most challenging. NVQWOA approaches the problem from a different angle, relying on exploitation of quantum mechanics and combinatorial reasoning rather than pattern recognition, requiring only the information necessary to define the optimisation task. In the longer term, hybrid approaches that combine GNN-derived feature extraction with quantum optimisation could exploit the strengths of both approaches, potentially yielding superior performance on large-scale, unlabelled graph similarity tasks.

The aim of this thesis is to investigate the performance of the non-variational quantum walk-based optimisation algorithm on the unlabelled graph similarity problem. Specifically, it seeks to characterise how NVQWOA redistributes probability amplitude toward optimal and near-optimal solutions, how its performance depends on hyperparameter choices and objective functions, and how the structure of the solution space influences its optimisation dynamics. The study also examines how distance measures, particularly subshell distance, reveal the relationship between solution quality and structural proximity.

This investigation is carried out through classical simulations of NVQWOA. The results are analysed to assess the algorithm's amplification behaviour, hyperparameter dependence, and sensitivity to solution structure. Through this work, the thesis aims to evaluate NVQWOA's potential as a practical tool for quantum optimisation and to lay the foundation for future research into structure-aware quantum algorithms for combinatorial problems.

1.1 Thesis Outline

- Chapter 2: Theory starts by outlining combinatorial optimisation and the graph similarity problem, then introduces the fundamentals of quantum computation, including variational quantum algorithms and the non-variational quantum walk-based optimisation algorithm.
- Chapter 3: Methods outlines the formulation of the NVQWOA simulation, describing the software tools used and the initial conditions selected.
- Chapter 4: Results and Discussion analyses the results obtained from the NVQWOA simulation and draws conclusions on suitability, performance, and scalability.
- This is followed by Chapter 5, outlining future work, and chapter 6, concluding the thesis.

Chapter 2

Theory

2.1 Combinatorial Optimisation Problems

Combinatorial optimisation problems consist of a discrete set of feasible solutions, which can be represented as a set S, with finite order $N = |S|$. Each of the N solutions $\mathbf{x} \in S$ can be represented as a vector of n variables $\mathbf{x} = (x_1, x_2, ..., x_n)$, where $x_j \in \{0, 1, ..., k-1\}$ are called the *decision variables*. The solution space S may contain all possible solutions of this form (of which there are $N = k^n$) or S may be restricted to solutions which satisfy some constraint (such as in the case of permutation-based problems where $k = n$ and $N = n!$). Each solution has an associated objective value (or cost), given by a objective function (or cost function) $f : S \to \mathbb{R}$, which gives a measure of the quality of any solution. The goal of the optimisation problem is to find the *optimal solution* $\mathbf{x}^*$ that maximises (or minimises) the objective (or cost) function:

$$\mathbf{x}^* = \arg\min_{\mathbf{x} \in S} f(\mathbf{x}) \text{ or } \mathbf{x}^* = \arg\max_{\mathbf{x} \in S} f(\mathbf{x})$$

The structure of the objective function is determined by the problem definition. The problem definition consists of a series of criteria which are defined on a subset of the variables in $\mathbf{x}$, and the objective function conveys the extent to which a solution satisfies the criteria.

Combinatorial optimisation problems are often NP-hard, meaning that no known polynomial-time algorithm can exactly solve all instances of the problem. Examples of NP-hard combinatorial optimisation problems include the travelling salesman problem [18] and the knapsack problem [19].

The main challenge in solving these problems is that they often require a near-exhaustive search of the solution space, which usually grows exponentially or super-exponentially with problem size. An example of this super-exponential growth will be present in the graph similarity problem in section 2.2.

Research in combinatorial optimisation algorithms showing exponential speedup includes [20]:

- Exact algorithms that find optimal solutions for special problem instances with certain properties.
- Approximate algorithms that find solutions which are near the optimal solutions for general problem instances.

Approximate algorithms are the focus of this thesis, as they can be applied to a wide range of practical problems, and are an area of active research in quantum computing [1, 21].

2.2 Graph Similarity Problem

The graph similarity problem involves determining how similar two graphs are to each other. This can be useful in various applications, such as social network analysis, bio-informatics, and pattern recognition [22, 6, 23].

There is a need to distinguish between the graph similarity problem and the graph isomorphism problem.

A pair of graphs G_1 and G_2 are considered isomorphic ($G_1 \cong G_2$) if and only if there exists a bijection between their vertex sets such that any two vertices in G_1 are adjacent if and only if their corresponding vertices in G_2 are also adjacent. In this definition, the graphs are understood to be undirected, non-labelled, non-weighted graphs. However, isomorphism may apply to all other notions of graph by adding requirements to preserve any additional structures.

The graph isomorphism problem is then the task of correctly creating that bijection (or equivalently, proving that one exists). Algorithms that solve this problem are trivially verifiable, as checking the correctness of the solution only involves checking that the vertex bijection correctly preserves the edges. Since the graph isomorphism problem can be posed as a 'yes-no' question on a set of inputs values, it is called a decision problem.

In contrast, the graph similarity problem is the problem of calculating how close a pair of graphs are to being isomorphic. This cannot be posed as a 'yes-no' problem, as many magnitudes of similarity must meaningfully captured. There are multiple measures of graph similarity. Some examples include:

- *Graph Edit Distance (GED):* Counts how many 'edits' are required to transform one graph into another, where 'edits' include inserting/deleting a vertex or edge, or substituting one vertex or edge with another. The lower the distance, the more similar the two graphs are [8].
- *Maximum Common Subgraph (MCS):* Finds the largest induced subgraph that is common to the two given graphs. The larger the subgraph, the more similar the two graphs are.
- *Frobenius Distance:* Calculates the distance between two graphs using the difference between their adjacency matrices. The lower the distance, the more similar the two

graphs are.

- *Edge Overlap (EO):* Finds the proportion of matching entries in the adjacency matrices of the graphs. The greater the proportion, the more similar the two graphs are.[1]

The choice of measure often depends on the specific application and the properties of the graphs being compared. Considerations such as cost, time, and interpretability inform which measure is best for a practical problem.

Koutra et al. [24] formalised a set of axioms that graph similarity measures should conform to:

1. Identity: $sim(G_1, G_2) = 1 \iff G_1 \cong G_2$
2. Symmetric: $sim(G_1, G_2) = sim(G_2, G_1)$
3. Zero: For a clique graph G_1 and a empty graph G_2, $\lim_{n\to\infty} sim(G_1, G_2) = 0$

The measures listed above can all be adapted to satisfy these axioms. For example, Koutra's axioms naturally require that the output of a graph similarity function remain bounded in the interval [0, 1]. To adapt a measures of distance such as GED to this requirement this can be accomplished by taking the inverse exponential of the distance, $e^{-f(G_1,G_2)}$. Other measures, like EO, trivially adhere to these axioms.

Most graph similarity measures assume a given mapping between the vertices of the graphs. Unlike in graph isomorphism, this vertex mapping doesn't need to be a bijection, since there exists a notion of similarity between graphs with a different number of vertices. For graph similarity methods such as those listed previously, the choice of vertex mapping can change the final result. However, Koutra's axiom 1 requires that graph similarity measures can identify isomorphic graphs by giving them a similarity of 1.

So to satisfy Koutra's axioms, we must find a mapping between the graphs that maximises their similarity.

To find the best mapping between labelled graphs, one can simply 'pair up' the vertices in the graphs so that each pair consists of one vertex from each graph that share the same label. This significantly reduces the number of possible mappings (often to just 1), so finding the mapping with the maximum similarity is simple.

To find the best mapping between unlabelled graphs is very challenging. In the general case, the similarity associated with every possible permutation of mappings must be checked until either a mapping that shows isomorphism (or another verifiable global maximum) is found, or the entire space of mappings has been searched. An example of vertex mappings is shown in Figure 2.1, which shows how edge overlap (EO) and graph edit distance (GED) are strongly affected by the choice of a vertex mapping.

As such, the unlabelled graph similarity problem can be defined in terms of a combinatorial optimisation problem.

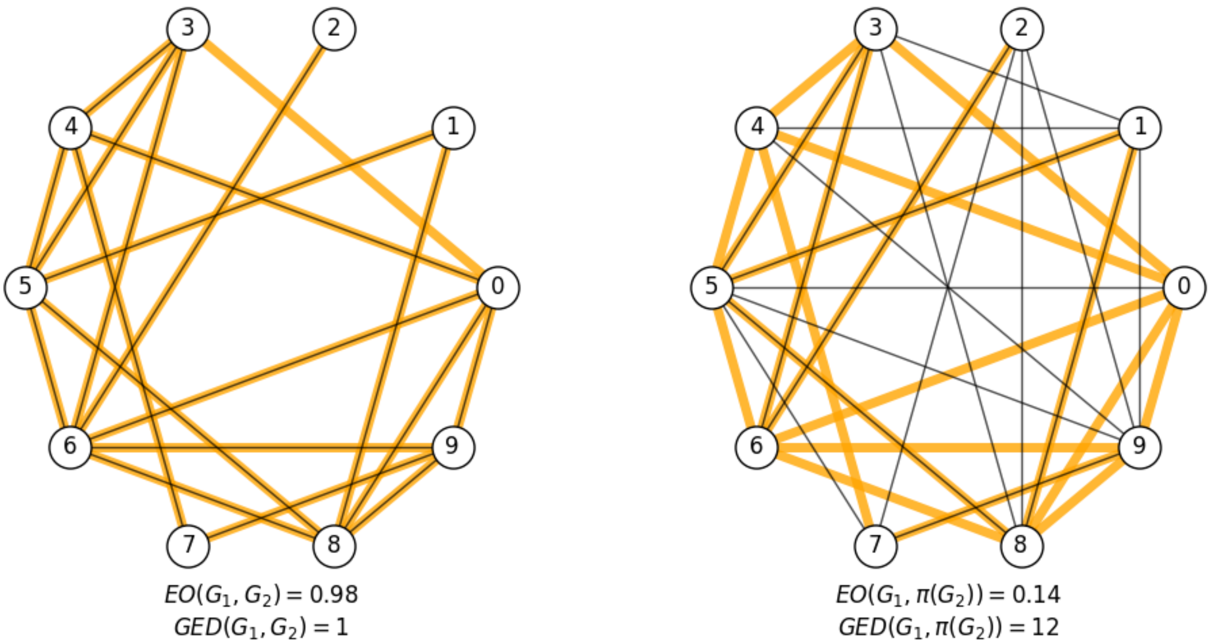


Figure 2.1: Diagram depicting how the choice of vertex mapping affects the similarity between G_1 (orange) and G_2 (black). (Left) depicts the case when the two graphs are maximally aligned, (right) depicts a random case with misalignment. The network structure of G_1 and G_2 were not modified, but an alternate vertex mapping π was applied to G_2 with respect to G_1.

2.2.1 Problem Definition

A formal definition for the unlabelled graph similarity problem is as follows:

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two finite, simple graphs and considered *unlabelled* in the sense that vertex identifiers carry no intrinsic meaning.

Let $\Pi(V_1, V_2)$ denote the set of all maps $\pi : V_1 \to V_2$.

Given a base similarity function $f : \mathcal{G} \times \mathcal{G} \to \mathbb{R}$ that compares labelled graphs from the class $\mathcal{G}$ of all finite simple graphs, the *unlabelled similarity* between G_1 and G_2 is then defined as

$$\mathrm{Sim}(G_1, G_2) = \max_{\pi \in \Pi(V_1, V_2)} f(G_1, \pi(G_2)),$$

where $\pi(G_2)$ denotes the relabelling of G_2 induced by π.

The unlabelled graph similarity problem is the problem of computing $\mathrm{Sim}(G_1, G_2)$. Since Π is set of maps between two finite sets, the size of the set $|\Pi|$ scales super-exponentially. In the case that $|V_1| = |V_2|$ and the maps π denote permutations of n vertices (), $|\Pi| = n!$.

This problem is known to be NP-hard, meaning that there is no known polynomial-time algorithm to solve it for all instances. As a result, various heuristic and approximation techniques have been developed to tackle the graph similarity problem.

The primary challenge is the large solution space of the problem, which may be addressed by quantum computers, as the amount of resources required to encode all solutions would grow only linearly or as a low order polynomial. This thesis will explore an approximate quantum algorithm, described in the remainder of this chapter.

2.3 Quantum Computing

Quantum computing is based on the principles of quantum mechanics, which describe the behaviour of particles at the atomic and subatomic levels.

This section provides a brief overview of the fundamental concepts of quantum computing that are relevant to understanding the non-variational QWOA and its application to the graph similarity problem.

2.3.1 Qubits and Quantum Superposition

The fundamental unit of information in classical computing is the bit, which can exist in one of two states, typically written as 0 or 1.

In quantum computing, the basic unit of information is the quantum bit or qubit. Unlike classical bits, which can only be in one state at a time, qubits can exist in a superposition of states.

Qubits are physically realised with two-state quantum systems, such as the spin of an electron in which the two states can be taken as spin up or spin down. Qubits are mathematically described by their two-dimensional complex state vectors. Systems of qubits are described in a 2^n dimensional complex vector space, where n is the number of qubits. A qubit's state (or the state of a collection of qubits) can be represented as the complex weighted sum of its basis states. A common choice of basis states is called the standard or computational basis, consisting of the positive and negative eigenvalues of the Pauli Z operator, commonly notated as $|0\rangle$ and $|1\rangle$, respectively. A general state of a single qubit can be expressed in the standard basis as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where α and β are complex numbers that satisfy the normalisation condition $|\alpha|^2 + |\beta|^2 = 1$. The Born rule states that the coefficients $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of measuring the qubit in the states $|0\rangle$ and $|1\rangle$, respectively.

When multiple qubits are combined, their joint state is represented by the tensor product of their individual states. For example, the state of a two-qubit system can be expressed as:

$$\begin{aligned}|\psi_{12}\rangle &= |\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \\ &= \alpha_1\alpha_2 |0\rangle |0\rangle + \alpha_1\beta_2 |0\rangle |1\rangle + \beta_1\alpha_2 |1\rangle |0\rangle + \beta_1\beta_2 |1\rangle |1\rangle \\ &= \alpha_1\alpha_2 |00\rangle + \alpha_1\beta_2 |01\rangle + \beta_1\alpha_2 |10\rangle + \beta_1\beta_2 |11\rangle\end{aligned}$$

where $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ are the basis states of the two-qubit system.

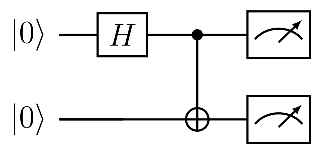
This allows a quantum computer with n qubits to represent 2^n states simultaneously, which is a key feature that enables quantum parallelism, allowing quantum computers to perform certain computations more efficiently than classical computers.

Despite this parallelism, the computational power is limited by the fact that measuring a quantum state collapses it to one of its basis states, yielding only a single outcome. Therefore, quantum algorithms must be carefully designed to manipulate the amplitudes of the quantum states such that the desired outcomes have higher probabilities of being measured.

2.3.2 Quantum Gates, Entanglement

Quantum gates are the building blocks of quantum circuits, analogous to classical logic gates. They manipulate the states of qubits through unitary transformations, which are reversible operations that preserve the total probability of the quantum system. Common quantum gates include the Hadamard gate, which creates superposition, and the CNOT gate, which entangles two qubits.

We can visually represent a series of quantum gates acting on qubits using a quantum circuit diagram. For example, the following diagram shows a simple quantum circuit with two qubits and two gates:



This diagram represents a circuit where the first qubit is put into superposition by the Hadamard gate (H), and then a CNOT gate acts on the second qubit depending on whether the first qubit is in a $|1\rangle$ state or the $|0\rangle$ state. Finally, both qubits are measured.

The quantum state begins as:

$$|\psi\rangle = |00\rangle$$

Then after the Hadamard gate is applied, the quantum state is:

$$|\psi_H\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

Finally the CNOT gate is applied and the quantum state is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The measurements can be performed relative to any basis in the qubit's vector space. Commonly, measurements are performed in the same basis as that is used to describe the qubits.

This state is an example of entanglement, a unique quantum phenomenon where the states of two or more qubits become correlated such that the state of one qubit cannot be described independently of the state of the other qubit(s). For example, in the entangled state above, if one were to measure the first qubit to be in the state $|0\rangle$, that necessarily requires the second qubit to also be in the state $|0\rangle$.

The measurement of one of the entangled qubits has collapsed the superposition for both qubits. This surprising result was initially deemed unphysical on the grounds that it violates local realism by Einstein, Poldowsky, and Rosen, the first to describe this behaviour [25]. Nevertheless, quantum entanglement and its implications have been repeatedly verified through experimentally violating Bell's inequality [26], establishing that the correlations produced from quantum entanglement cannot be explained by local hidden variables [27, 28, 29]. These correlations are fundamental to quantum algorithms. Without entanglement, quantum computers could be efficiently simulated by classical computers, negating any advantage [30].

2.3.3 Quantum Algorithms and Challenges

Quantum algorithms seek to leverage properties of qubits and quantum gates to solve specific problems more efficiently than classical algorithms. Notable examples include Shor's algorithm for integer factorisation [11], which runs exponentially faster than the best-known classical algorithms, and Grover's algorithm for unstructured search [10], which provides a quadratic speedup over classical search algorithms and is the optimal quantum algorithm for unstructured searches [31].

Large, universal, fault-tolerant quantum computers that can run these algorithms at useful scales will require large numbers of qubits. While researchers have made significant progress in error correction, coherence times, qubit connectivity, and gate fidelities, fault tolerant quantum computing may take decades to achieve. In the meantime, Noisy Intermediate-Scale Quantum (NISQ) computers already exist, and can run small-scale quantum algorithms that can provide insights into quantum behaviour and inform the development of larger systems [32].

These NISQ devices are limited by the number of qubits, gate fidelities, and coherence times. As a result, quantum algorithms designed for NISQ devices must be efficient in terms of qubit usage and circuit depth to minimise the impact of noise and errors [32].

Even if NISQ devices cannot solve problems faster than the best classical computers using the best classical algorithms (particularly with the emergence of exascale supercomputing), they could still provide an economic advantage by using less energy or by being lower cost to build and operate [32].

NISQ algorithms are designed to work within these low-resource constraints. These algorithms typically involve a hybrid approach, where a quantum computer is used to prepare and measure quantum states, while a classical computer optimises parameters based on the measurement results. Some examples of NISQ-era algorithms are described in the following sections.

2.4 Variational Quantum Algorithms

Variational Quantum Algorithms (VQAs) are a class of hybrid quantum-classical algorithms that leverage the strengths of both quantum and classical computing to solve optimisation problems. They are particularly well-suited for NISQ devices due to their relatively low resource requirements and robustness to noise. The goal of a VQA is to learn the parameters for a quantum circuit which evolves an initial state into a final state that encodes the computational objective.

2.4.1 VQAs in General

VQAs implement a parametrised quantum circuit (PQC) whose unitary evolution is governed by a set of d classically tunable parameters $\theta \in \mathbb{R}^d$. The quantum device prepares the state

$$|\psi(\theta)\rangle = U(\theta)\,|0\rangle^{\otimes n}$$

where $U(\theta)$ is a depth-restricted circuit composed of parameterised single- and multi-qubit gates, and n is the number of qubits in the input to the quantum computer. Commonly, $d = \mathcal{O}(\mathrm{poly}(n))$, since we seek efficient circuits that are polynomial in resources [15].

The objective function of the algorithm is typically expressed as the expectation value of a Hermitian operator H with respect to the variational state:

$$C(\theta) = \langle\psi(\theta)|\,H\,|\psi(\theta)\rangle\,.$$

This cost function is estimated on the quantum device via repeated projective measurements in an appropriate basis. A classical optimiser updates θ with the aim of minimising (or maximising) $C(\theta)$, thus driving the quantum circuit towards a state that approximates the optimal configuration with respect to the target problem. The algorithm repeatedly prepares this variational quantum state, measures its expectation value, and optimises the parameters until a certain number of iterations is complete or until measurements of the quantum state have an approximation ratio sufficiently close to 1.

To summarise, VQAs consist of a two-step loop, pictured in Figure 2.2: a quantum computer evolves the initial state to a final state via a PQC, then a classical computer measures the

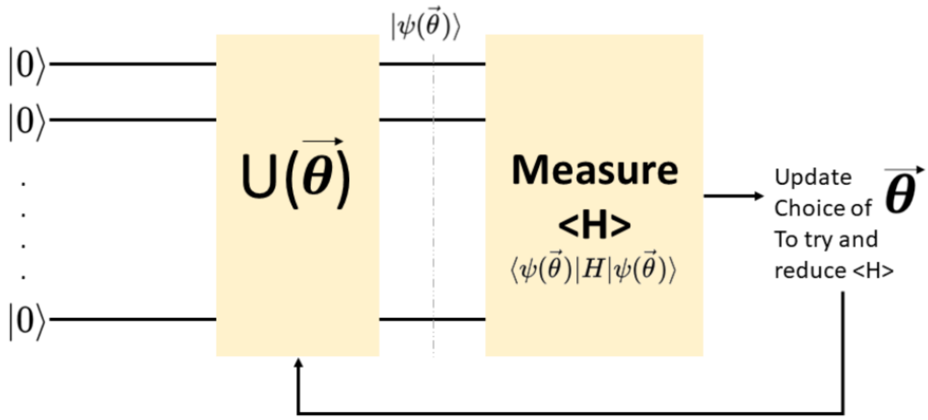


Figure 2.2: Schematic diagram of a variational quantum algorithm structure. The yellow boxes are operations performed on a quantum computer

expectation value of the final state and uses this data to update the PQC. At the end of this loop, the quantum computer's output is now a good approximation of the optimal configuration.

2.4.2 QAOA

The Quantum Approximate Optimisation Algorithm (QAOA) is a type of VQA designed to solve combinatorial optimisation problems, introduced by Farhi et al. [12] In this formulation, the PQC $U(\theta)$ is defined by alternating applications of unitaries. One is a quality-based phase separation unitary $U(C,\gamma)$ associated with the problem instance and the other is a mixer $U(B,\beta)$ that induces interference.

We are given some combinatorial optimisation problem, specified by m clauses and solutions $\mathbf{x} = (x_1, x_2, ..., x_n)$ that have binary decision variables $x_j \in \{0,1\}$. Each clause is a constraint on a subset of the decision variables which is satisfied for certain assignments of those bits and unsatisfied for other assignments. The objective function is the number of satisfied clauses,

$$C(\mathbf{x}) = \sum_{\alpha=1}^{m} C_\alpha(\mathbf{x})$$

where $\mathbf{x} =$ is encoded as a bit string and $C_\alpha(\mathbf{x}) = 1$ if $\mathbf{x}$ satisfies clause α and 0 otherwise.

The initial state for the QAOA is an equal superposition state over the computational basis states:

$$|s\rangle = \frac{1}{\sqrt{2^n}} \sum_{\mathbf{x}\in S} |\mathbf{x}\rangle$$

Each qubit encode both choices for the binary decision variable for its register, so all the qubits taken together form one 'qubit string' that encodes the complete solution space S.

Define a unitary operator $U(C,\gamma)$ which depends on an angle $\gamma \in [0, 2\pi]$:

$$U(C,\gamma) = e^{i\gamma C} = \prod_{\alpha=1}^{m} e^{-i\gamma C_{\alpha}},$$

which is the phase separation unitary.

Define the operator B which is the sum of all single bit Pauli X operators applied to the jth register,

$$B = \sum_{j=1}^{n} \sigma_j^x.$$

Finally, define the $U(B,\beta)$ operator which depends on the parameter $\beta \in [0,\pi]$:

$$U(B,\beta) = e^{-i\beta B} = \prod_{j=1}^{n} e^{-i\beta\sigma_j^x},$$

which is the mixing unitary. which induces interference between the qubits. Other mixers are introduced for use in the NVQWOA in section 2.6.

Now, for any integer $p \geq 1$, and $2p$ angles $\gamma = (\gamma_1, ..., \gamma_p)$ and $\beta = (\beta_1, ..., \beta_p)$ we define the final quantum state:

$$|\gamma,\beta\rangle = U(B,\beta_p)U(C,\gamma_p)\dots U(B,\beta_1)U(C,\gamma_1)\,|s\rangle$$

This quantum state is acquired by repeatedly applying $U(C,\gamma)$ then $U(B,\beta)$. The action that $U(C,\gamma)$ has on each individual state is to apply a local phase shift to its complex amplitude proportional to the quality of the solution represented by that state. Because this results in the phases of the solutions being separated from one another, this unitary is called the phase separation unitary. $U(B,\beta)$ induces interference between the complex amplitudes, and is often called a transverse-field Hamiltonian, because it is defined as the sum of Pauli X operators that act transversely to the computational basis defined by Pauli Z operators. This allows all computational basis states to be reached from any other via the repeated application of $U(B,\beta)$. This mixing action caused by $U(B,\beta)$ is equivalent to a continuous-time quantum walk (CTQW) on a graph, with solutions as the vertices of the graph and the edges connecting solutions that differ by only one binary decision variable, i.e. a hypercube in n dimensions [16].

After obtaining the final quantum state $|\gamma,\beta\rangle$ we can now do the usual VQA method of taking several shots to measure the expectation value, optimising the parameters γ and β according to our results, and repeating to obtain good approximations of the optimal solution. This process is pictured in Figure 2.3.

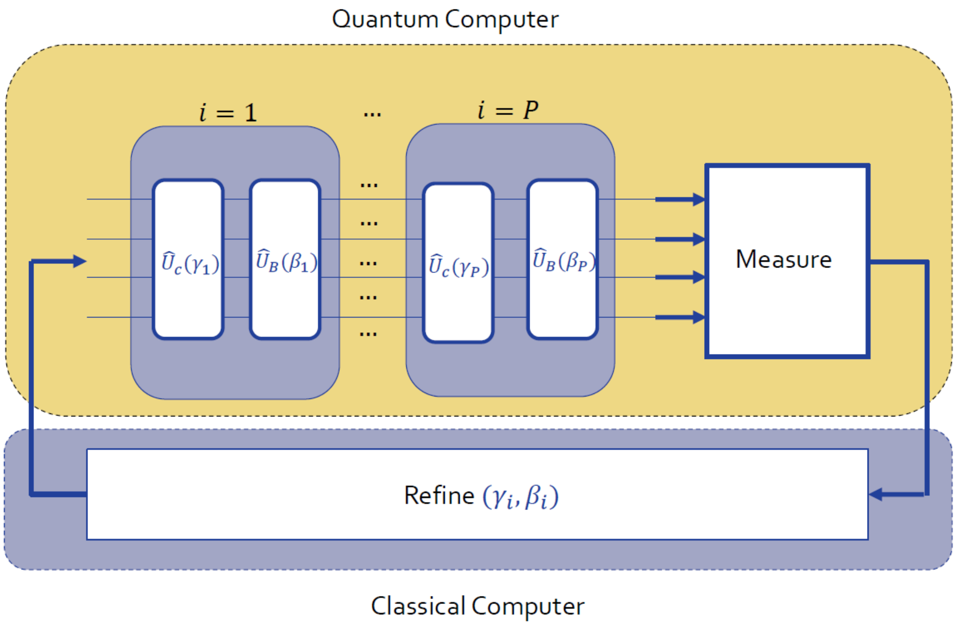


Figure 2.3: Schematic of the QAOA algorithm [1]

2.5 Challenges and Gaps in Literature

Applying the QAOA to the graph similarity problem is challenging, particularly because the QAOA encodes solutions as bitstrings that can be any combination of binary values. [12, 13, 14]. A natural scheme to encode any possible vertex mapping for two graphs, G_1 and G_2, each with n vertices requires n^2 bits (each bit represents one of n vertices in G_1 mapping to one of n possible vertices in G_2). This implies a possible solution space with $|S| = 2^{n^2}$. However, the solutions to graph similarity problems take the form of permutations, where each vertex must map to a unique partner, so the size of the feasible solution space is $|S| = n!$. The form of the solutions being constrained this way means the vast majority of solutions will be unfeasible ($n! \ll 2^{n^2}$).

To deal with unfeasible solutions in the QAOA, most studies include penalty terms into their objective functions that reduce the quality of unfeasible solutions [18, 33]. While effective in principle, this approach increases the complexity of the cost landscape and the number of parameters to be optimised, adding to the numerical overhead. Pritchard et. al applied the QAOA to the graph similarity problem using a novel encoding that reduced the number of required qubits [1]. Although amplification of optimal solutions was demonstrated, achieving strong performance required global parameter optimisation involving thousands of cost function evaluations, even for relatively shallow circuits with $p \in [1, 6]$. These computational demands limited the study to graphs with $n \leq 9$.

More broadly, the effectiveness of VQAS including QAOA relies on the computationally expensive optimisation of often large sets of variational parameters [14]. Their performance is highly dependent on the initial choices of parameters prior to training [13]. As Bennett et. al argues, VQAs can end up replacing one difficult optimisation problem with another [16].

This motivates the search for alternative quantum optimisation approaches that retain strong performance while reducing classical optimisation overhead. One such approach is the non-variational quantum walk-based optimisation algorithm (NVQWOA), which is described in the following section.

2.6 Non-Variational Quantum Walk-based Optimisation Algorithm

The NVQWOA is a novel algorithm introduced by Bennett et al. that builds upon and addresses the variational challenges of the QAOA [16, 2]. Like the QAOA, it alternates between applying two unitaries: a phase-separation unitary that encodes the objective function into the complex phase of the states and a mixing unitary that induces quantum interference between the solutions. Unlike QAOA or other VQAs, the parameters defining the unitaries are not variationally tuned by a classical optimiser. Instead, they are predetermined by the choice of only three hyperparameters γ, t, β. Additionally, the structure of the mixer is chosen to match the structure of the problem, to connect nearest-neighbour solutions to one another.

Simulations have demonstrated that this non-variational approach yields rapid convergence to global optima across diverse problem types, including MaxCut, maximum independent set, k-means clustering, capacitated facility location, and the quadratic assignment problem, using only modest circuit depths and a handful of iterations [2].

2.6.1 Algorithm Description

Using the terms and definitions of combinatorial optimisation problems from Section 2.1:

The algorithm begins with an initial superposition over all possible solution states

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{x} \in S} |\mathbf{x}\rangle .$$

Each iteration of the NVQWOA applies two unitaries

$$U_Q(\gamma) = e^{-i\gamma Q}, \;\; U_M(t) = e^{-itA},$$

where Q is a diagonal operator such that $Q\,|\mathbf{x}\rangle = f(\mathbf{x})\,|\mathbf{x}\rangle$ encodes the objective function values, and A is the adjacency matrix of a mixing graph whose vertices correspond with feasible solutions and whose edges connect nearest-neighbour solutions to one another.

The phase-separation unitary applies a phase shift to each solution state $|\mathbf{x}\rangle$ proportional to the objective value of that solution $f(\mathbf{x})$, while the mixing unitary $U_M(t)$ implements a continuous time quantum walk (CTQW) on the mixing graph, redistributing complex amplitudes through neighbouring solutions. Repeated alternating application of these unitaries produces a designed interference process in which amplitudes associated with optimal or near-optimal solutions interfere constructively, amplifying their measurement probabilities.

The final quantum state after p iterations of each unitary is

$$|\gamma, t, \beta\rangle = \left[\prod_{i=0}^{p-1} U_M(t_i) U_Q\left(\frac{\pm\gamma_i}{\sigma}\right)\right] |s\rangle \,,$$

where the $\pm$ accounts for whether the goal is minimisation $(-)$ or maximisation $(+)$, σ is the standard deviation of the initial objective function landscape (which can be efficiently approximated through random sampling of $p = 0$), t_i and γ_i are obtained from specific formulae in terms of $\gamma, t,$ and β and where $\gamma > 0, t > 0$, and $0 < \beta < 1$:

$$\gamma_i = \left(\beta + (1-\beta)\frac{i}{p-1}\right)\gamma, \;\; t_i = \left(1 - (1-\beta)\frac{i}{p-1}\right)t,$$

so the angles γ_i increase linearly with i and the mixing times t_i decrease linearly with i.

2.6.2 Mixing Unitary

A key conceptual advance of the NVQWOA lies in the explicit construction of a problem-structured mixing graph (also referred to as mixing unitary or mixer). The vertices of the graph represent feasible solutions, and edges connect nearest neighbour configurations, typically those separated by a minimal Hamming distance or by a single elementary move. The resulting adjacency matrix A defines a vertex-transitive graph whose degree d is polynomial in problem size.

Because the mixer only connects the feasible solutions to one another, complex amplitudes are only distributed across feasible solution states, so the NVQWOA is able to avoid most of the problems caused by unfeasible solutions. For certain problems where the unfeasibility is not due to the inherent structure of the solution, penalty terms may still need to be added [16, 2].

2.6.2.1 Mixer Examples

In his work, Bennett provides some examples of mixing graphs with known efficient implementations on a quantum computer [16, 2]:

Binary Mixer: For problems with n binary decision variables $x_j \in \{0, 1\}$, the mixing graph is a n-dimensional hypercube, which is identical to the transverse-field Hamiltonian from the QAOA. Solutions are connected to the n other solutions which differ by one bit-flip.

Integer Mixer: For problems with n integer decision variables $x_j \in \{0, 1, ...k-1\}$. Solutions are connected to the $n(k-1)$ other solutions which differ in the choice of one decision variable.

Permutation Mixer: For problems with n integer decision variables $x_j \in \{0, 1, ...k-1\}$ where the solutions are constrained to not have repeated decision variables. Solutions are connected to the $\frac{1}{2}n(n-1)$ other solutions which are transpositions of each other.

The choice of mixing graph directly determines how probability amplitudes are propagated, making it possible to exploit problem structure, unlike the transverse-Hamiltonian mixer in the QAOA, which applies the same mixer to all combinatorial optimisation problems[12].

2.6.2.2 Mixer Subshells

The distance $dist(u, v)$ between two vertices u, v on a graph is the minimum length of the paths connecting them. A subshell is a subset of the vertices of the mixing graph, defined in relation to a reference vertex u. The subshell h_u contains all vertices that are a distance h away from u. The set of all subshells for a graph form equivalence classes on the set of the vertices of the graph. An example of the subshell structure for a small mixer is shown in Figure 2.4

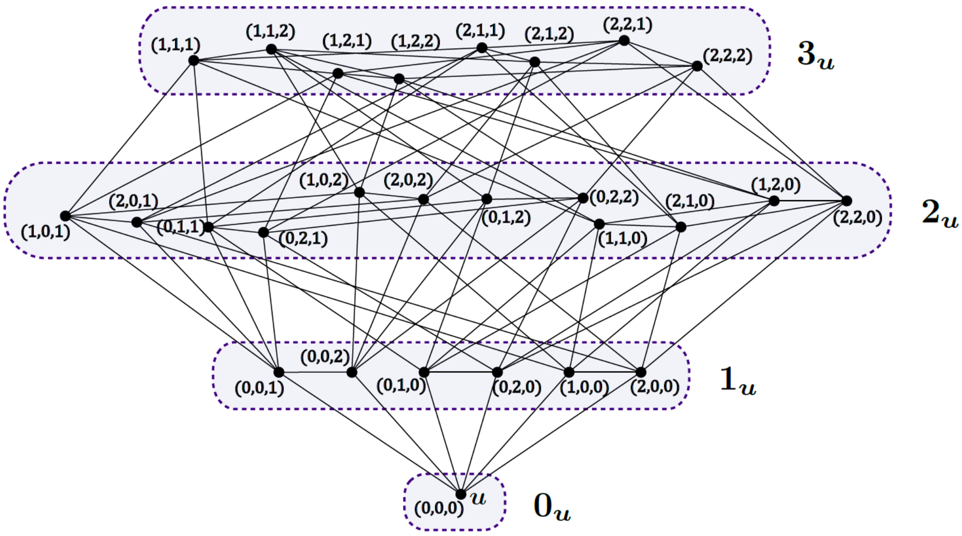


Figure 2.4: Example integer mixer for a problem with $n = 3, k = 3$. The graph is partitioned into subsets $h_{(0,0,0)}$ for $h = 0, 1, 2, 3$. [2]

For the NVQWOA to perform well on a combinatorial optimisation problem, it is expected to satisfy two conditions. First, the action of the mixer must cause probability amplitude from each vertex to be distributed to other vertices such that the complex phase of the distributed probability amplitudes are proportional to their distance on the mixing graph from the initial vertex [16, 2].

The second condition is that the following relationship should be at least approximately satisfied:

$$(\mu_{h\mathbf{x}} - f(\mathbf{x})) \approx -m_h(f(\mathbf{x}) - \mu),$$

where $\mu_{h\mathbf{x}}$ is defined as the mean objective function value of solutions contained in $h_{\mathbf{x}}$, μ is the mean objective function value of all solutions in S, and m_h is a positive constant. Also, the proportionality m_h should increase monotonically with increasing h (up to a distance which is a considerable fraction of the graph's diameter D).

Chapter 3

Methods

We simulated the application of the NVQWOA to the graph similarity problem. The measure of graph similarity chosen was Edge Overlap, as it is a low-cost measure that still serves as a interpretable measure of similarity. The aim was to maximise this objective function through matrix exponentiation and multiplication,analytically computing the manipulation of state vector such that the states which corresponded with the optimal solutions (those with the greatest similarity) had an amplified probability of being measured.

There were two phases to the analysis. First, we applied the algorithm to one problem at a time and small problem sizes up to $n = 9$, using standard matrix exponentiation. Then we applied the algorithm to several problem instances simultaneously and had problem sizes up to $n = 10$.

3.1 Simulation

To investigate the performance of the NVQWOA on graph similarity problems, we chose to perform classical simulations of a quantum computer, rather than directly performing the algorithm on a quantum computer. Simulation allows for computation with negligible stochastic or decoherence errors and complete analysis of the evolving quantum state, something which is inaccessible on real quantum hardware but useful for characterising performance and benchmarking.

Simulation of quantum algorithms and computers can be approached in many different ways. Tensor networks simplify circuits into lower-entanglement systems but has exponentially degrading performance as circuit depth and entanglement entropy. Stabiliser-based approaches can efficiently simulate Clifford-only circuits and certain restricted non-Clifford extensions but cannot capture the parameterised unitaries used in the NVQWOA. State vector simulation computes the full evolution of the state vector, but can be costly in computational resources.

State vector simulation was chosen as the strategy for its accuracy. The Qiskit Aer simulator and related packages were not suitable as the permutation mixer required is a sparse operator, which is inefficient to simulate under the requirements of using tensor products of standard gates. Instead, we directly computed the relevant matrix operations. For the first phase of analysis, the matrix exponentiation and multiplication was able to be performed on local hardware using the well-known SciPy package. For the second phase of analysis, the large problem sizes required parallel computation. To implement this, we used QuOp, a Python framework for parallel simulation of quantum variational algorithms[34, 35]. QuOp provides efficient handling of large, sparse unitaries and supports MPI-based parallelisation, allowing simulation of high-dimensional state vectors across multiple compute nodes without loss of precision.

To simulate NVQWOA using QuOp, a parameter mapping was used to translate the hyper-parameters of the NVQWOA to the individual parameters of a QVA. The quantum circuit was defined to be the two unitaries of the NVQWOA; the phase separator U_Q, and the mixer U_M. With these modifications, QuOp could simulate the NVQWOA.

The parallelised simulations of NVQWOA were performed on the Setonix system at the Pawsey Supercomputing Centre.

3.2 Graph Similarity Measure

As described in section 2.2, there are many options for the choice of which graph similarity measure to use.

Graph edit distance is the most common choice, but is costly to compute, as it requires the equivalent of a pathfinding search, usually performed using the A^* search algorithm. Since computational resources were a significant pressure, we chose one of the least computationally expensive similarity measures to compute, the edge overlap. This also allows for comparison against Pritchard et al., who used edge overlap in their analysis of the QAOA applied to graph similarity [1].

To calculate the edge overlap, the number of matching entries in the adjacency matrices of the graphs are counted and the sum is divided by n^2, which corresponds to the size of the adjacency matrices and the maximum possible overlap. This simple calculation also makes the edge overlap highly interpretable, as one can imagine the vertex mapping as translating the vertices of one graph to the corresponding vertices in the other graph, and the edge overlap represents the proportion of edges that line up correctly. Edge overlap is well-defined for directed graphs, but directed graphs were not investigated in this thesis.

When the best vertex mapping is known between the two graphs, edge overlap satisfies Koutra's similarity measure axioms described in section 2.2, making it a 'good' measure of graph similarity.

A drawback of using edge overlap is that it is not well-defined for graphs with differing numbers of nodes, $|V_1| \neq |V_2|$. This restricts our investigation to graphs with an equal number of nodes.

The choice of graph similarity measure is not expected to have a strong effect on the observed results. The vertex mappings that give strong similarity with any particular measure will give strong similarity with most measures. For the choice of a graph similarity measure to affect the amplification, the measure would need to significantly alter which graphs were considered similar to one another, thereby changing the structure of the quality vector and phase-separation unitary.

3.3 Problem Instance Construction

There does not exist a universally used set of graph similarity problem instances. To test the algorithm on a wide group of fairly-chosen problems, we randomly generated the graphs using the Erdős-Rényi-Gilbert random graph model[36, 37].

The number of nodes n, was varied to investigate how changing problem sizes changed the performance of the algorithm $n = 8, 9, 10$ were used for problem instance generation. The probability of edge inclusion was fixed at 0.5 for simplicity and repeatability. At each value of n, 30 graphs were generated using the NetworkX Python library [38]. The seeds were selected starting from 0 and increasing by 1 for each successive problem instance. Any seeds that generated a graph isomorphic with any previously generated graphs would have been removed from the list of valid seeds. However, none of the seeds we used generated graphs isomorphic to one another, so the list of seeds was equivalent to the integers in [0,29].

Instead of calculating the similarity between the generated problem instances, the similarity was calculated between each graph and a copy of itself. This ensured similar distributions of similarity would arise between the different problem instances. For instance, the maximum similarity was always 1.

3.4 Phase Separator and Mixer Construction

The phase separating unitary was constructed in QuOp. For each vertex mapping, the quality (which we may refer to as edge overlap, similarity, or similarity score) was calculated and stored as an array. This array was converted into a diagonal unitary operator in QuOp.

Since we are considering graphs with an equal number of nodes n, vertex mappings match the arbitrary vertex indices of each graph to one another with some ordering, $\pi : \{1, ..., n\} \to \{1, ..., n\}$. (In the unlabelled graph problem, the vertices in each graph can still have indices, just not information that can inform how to match those indices together.) Each $\pi \in \Pi$ corresponds to a distinct reordering of the vertex indices. Therefore, vertex mappings can be equivalently written as permutations of n distinct decision variables.

From this, we can conclude that the correct choice of mixer is the permutation mixer, which connects solutions that are one swap of decision variables away from one another. Describing solutions in terms of vertex mappings, the permutation mixer connects solutions corresponding to mappings which differ only by two vertices that have had their targets swapped. Since these are the nearest neighbours to one another, we can also conclude that the minimum hamming distance from one solution to another is 2.

Bennett et al. proved that the permutation mixer satisfies the first NVQWOA condition, described in 2.6.2.2, and suggested approaches to simulating the permutation mixer [2].

We constructed the mixer by explicitly generating the sets of nearest-neighbours for each solution, storing the matrix as a column oriented array in Numpy [39] and converted to a compressed sparse row format for storage. This sparse matrix was loaded into QuOp using HDF5 parallel I/O operations[40].

3.5 Hyperparameter Optimisation

The NVQWOA has three hyperparameters: γ, t, β. Although sub-optimal parameters can produce quantum circuits that significantly amplify optimal and near-optimal solutions, it is expected that finding optimal values for these hyperparameters through three-dimensional numerical optimisation will give the best possible results.[2]

We used three different hyperparameter optimisation objective functions:

1. *Expected Value (EV)*: The expected value of the state vector after amplification. This would require the fewest shots to compute on real quantum hardware, and is a natural choice for optimisation [41].

2. *Optimal Solution Probability (OSP)*: The probability of measuring an optimal solution. This is impractical to measure on quantum hardware, as it requires foreknowledge of the optimal solution and an unfeasible number of shots. It is included in our investigation here to provide an upper bound by finding the best possible parameters.

3. *Conditional Value at Risk (CVaR)*: The expected value of the α-tail of the state vector. This takes more shots to measure on quantum hardware than the expected value, but has been shown to give faster convergence to better solutions[42].

Optimisation of the NVQWOA hyperparameters was done in QuOp during the NVQWOA simulation, using the Nelder-Mead method of numerical optimisation [43]. This method was chosen for its efficiency. Since it does not require the computation of derivatives, the method is faster to compute than other alternatives. One downside of using Nelder-Mead is that it can converge to a locally optimal set of hyperparameters, rather than a globally optimal set. Additionally, it does not have an in-built method to set boundaries on the possible parameters, so it is possible that it could converge to non-physical parameters such as $t < 0$.

3.6 Procedure

3.6.1 Amplification Analysis

For each of $n = 8, 9, 10$, the probability distribution for the different solutions was compared before and after the algorithm was applied. We expected that solutions with high similarity would be more likely to be measured after the algorithm was applied. Additionally, the probability to measure the optimal solution was calculated.

3.6.2 Solution Distance Analysis

To supplement the verification of amplification, it is useful to obtain statistics on the distance between solutions. This is because we expect that near-optimal solutions, which will have a low distance to the optimal solutions, should be amplified more than other solutions, which have a greater distance to the optimal solutions.

There are two useful notions of distance between solutions: subshell distance and Hamming distance, which are defined and explained below.

3.6.2.1 Subshell Distance

In this thesis, we use the term subshell distance to refer to the minimum distance on the mixing graph between a solution and an optimal solution. The term subshell in this notion of distance comes from the description in section 2.6.2.2, and is used to distinguish this distance measure from other approaches. The mixer we are using is the permutation mixer, so the subshell distance is defined as:

$$\mathrm{SD}(\mathbf{x}^1, \mathbf{x}^2) = \min\{m \in \mathbb{N}_0 : \exists \tau_1, ..., \tau_m \in \mathcal{T} \text{ with } \tau_m ... \tau_1 \mathbf{x}^2 = \mathbf{x}^1\}$$

where $\mathcal{T} = \{(ij) : 1 \leq i \leq j \leq n\}$ is the set of all transpositions in the symmetric group S_n and transpositions act by reindexing coordinates. For this analysis $\mathbf{x}^1$ is always an optimal solution. If there were multiple optimal solutions, the subshell distance from each optimal solution was found and the minimum value was used.

Subshell distance is a useful metric in the analysis because it directly represents how close the solutions are to each other on the mixing graph. By analysing how amplification is applied to solutions depending on their subshell distance, we can observe how probability is being moved by the mixing process. Additionally, if one were to perform a local search after finding a near-optimal solution, it would likely be a search of solutions with low subshell distance to the found solution, so analysing how the subshell distance of different solutions is relevant to the application of the NVQWOA.

3.6.2.2 Hamming Distance

Hamming distance is a simple yet powerful and highly interpretable measure for comparing objects. It measures the number of different decision variables for a solution. The Hamming distance for two solutions is defined as

$$\mathrm{HD}(\mathbf{x}^1, \mathbf{x}^2) = \sum_{i=0}^{n} \delta(x_i^1, x_i^2)$$

where δ is 1 if $x_i^1 \neq x_i^2$ and 0 otherwise. For this analysis $\mathbf{x}^1$ is always an optimal solution. If there were multiple optimal solutions, the Hamming distance from each optimal solution was found and the minimum value was used.

Hamming distance is a useful metric in the analysis because it gives a measure of distance that is not related to the structure of the algorithm and can be easily compared to other combinatorial optimisation problems and algorithms.

3.6.3 Subshell Analysis

Using the notion of subshell distance and the conditions described in section 2.6.2.2, we can characterise the suitability of this mixer to the problem of graph similarity.

To do so, we analysed the qualities of solutions within each subshell. The average difference in quality between the optimal solutions and the solutions in each subshell were found. This was called the mean quality gap. It is expected that solutions that have a high subshell distance are of lower quality, so the mean quality gap should ideally increase monotonically with subshell distance. If the mean quality gap was not monotonically increasing, then

The variance in quality within each subshell was found. This was called the subshell variance. The variance of the average variances was found. This was called the second-order variance.

3.6.4 Hyperparameter Analysis

The hyperparameters were selected by the optimisation procedure outlined in section 3.5, beginning with the initial parameters $\gamma = 1.0, t = 0.1, \beta = 0.4$. This set of initial parameters were recommended by Tavis et al. [16].

We stored the hyperparameters obtained for each problem instance with each optimisation objective, producing a set of nine hyperparameters for each problem instance.

Chapter 4

Results and Discussion

This chapter presents and interprets the results of classical simulations of the non-variational quantum walk-based optimisation algorithm (NVQWOA) applied to the unlabelled graph similarity problem. The primary objective is to evaluate the performance of NVQWOA in amplifying the probability of optimal and near-optimal solutions, to examine how its dynamics reflect the structure of the solution space, and to understand how its limited hyperparameter set interacts with different objective functions.

4.1 Single Graph Analysis

The first part of the analysis was performed on a randomly chosen graph. The seed 2025 was arbitrarily chosen to create a random graph by the method described in section 3.3. The NVQWOA was applied to this graph with fixed initial hyperparameters, $\gamma = 1.0, t = 0.1, \beta = 0.4$, in order to see if suboptimal hyperparameters can still produce amplification. The number of iterations was set to $p = 30$ for this phase of analysis. The results of this simulation are shown in Figures 4.1, 4.2, 4.3, and 4.4.

Figures 4.1 and 4.2 analyse the effect of the NVQWOA when applied to the graph $n = 9$, seed $= 2025$. The hyperparameters were set to $\gamma = 1.0, t = 0.1, \beta = 0.4$. From these results it is clear that probability amplitude has been channelled primarily into optimal and near-optimal solutions.

In Figure 4.1, the NVQWOA outperforms Grover's algorithm for an equivalent number of iterations. Since Grover's algorithm is optimal for searching an unstructured database [31], this suggests the the NVQWOA is successfully using the structure of the problem (as embedded in the phase separation and mixing unitaries) to amplify the optimal solutions. Another feature of note is that after $p = 15$ iterations, the OSP$=$ 0.0636607501 is at a maximum, and after this point, the OSP decreases with further iterations until it reaches 0.0342996063. The OSP appears to oscillate between those values, and ends on a value of 0.0506801642. This is caused by the phase-separation unitary U_Q mapping solution qualities to phase shifts. As discussed in

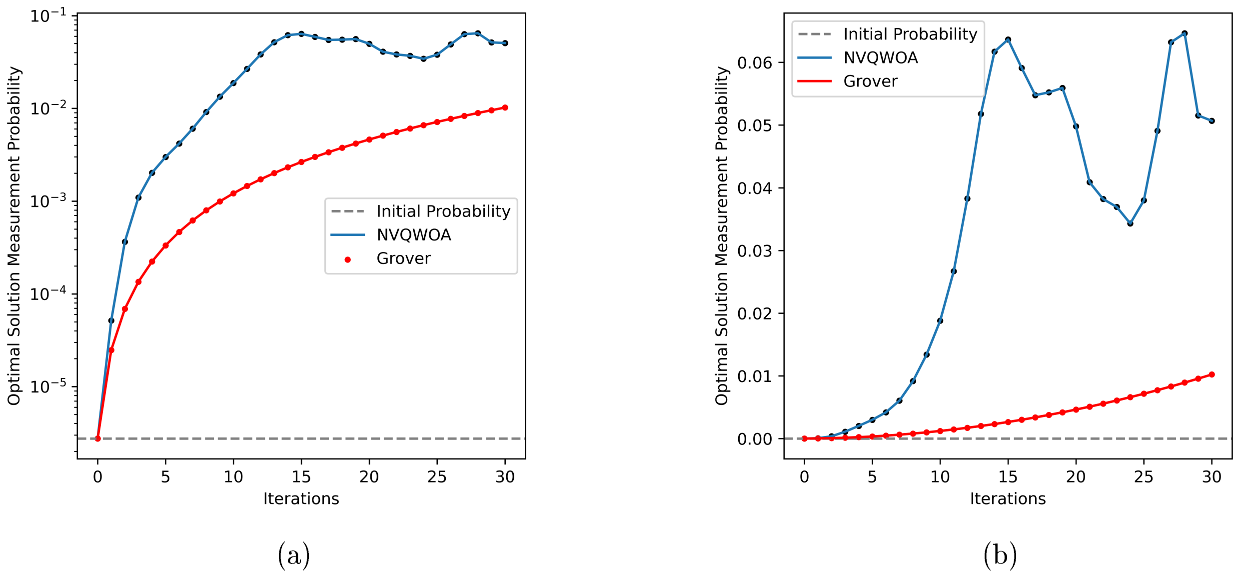


Figure 4.1: Simulation results for $p = 30$ NVQWOA applied to seed 2025 and Grover's algorithm for an unstructured database of equivalent size. Probability of measuring the optimal solution at each iteration. (a) log scaled y-axis, (b) linear y-axis.

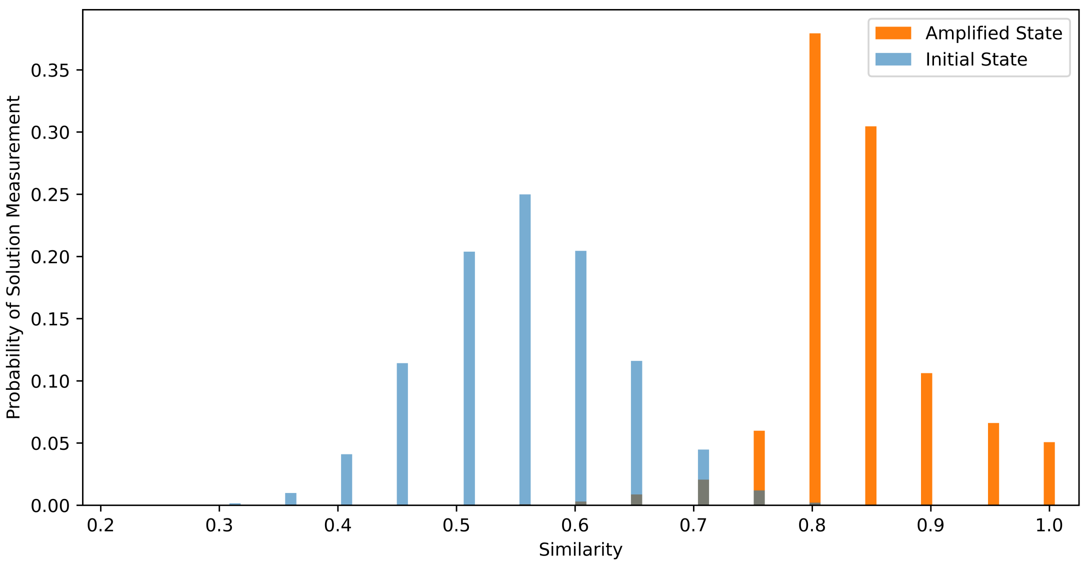


Figure 4.2: Simulation results for NVQWOA applied to seed 2025. Probability distributions for similarity as measured from the initial equal superposition state (blue) and the final amplified state (orange).

section 2.6, U_Q applies a local phase shift to each solution proportional to its quality. Repeated application of U_Q causes the phase of the solutions to rotate at different speeds relative to the number of iterations completed, so a solution with high quality may rotate its phase over 2π and be in phase with the low-quality solutions. The action of the mixer U_M will then cause attenuation of high-quality solutions. Afterwards, further application of U_Q rotates the phase of the high-quality solutions further, returning to a phase-separated form again, where U_M will successfully amplify the high-quality solutions. This effect explains the oscillation of the optimal solution probability. Over a large p, Bennett et al. comments that choosing smaller values for β and γ can preserve phase coherence at the cost of reducing optimal solution amplification, and that maximising total amplification over several iterations requires limiting amplification during initial iterations [2].

In Figure 4.2, the expected value for the similarity has increased from 0.556927 to 0.840676. The mode of the distribution has moved from a similarity of 0.56, which has 90690 states, to a similarity score of 0.80, which has only 748 states. Even though the probability of measuring the optimal solution has increased, it is not the new mode of the probability distribution. This is because there is only 1 optimal solution, so even though the other solutions have less amplification applied to them, they greatly outnumber the optimal solution usually by two or three orders of magnitude. Therefore, smaller probabilities that near-optimal solutions receive can sum up to be greater than the solution receiving the most amplification.

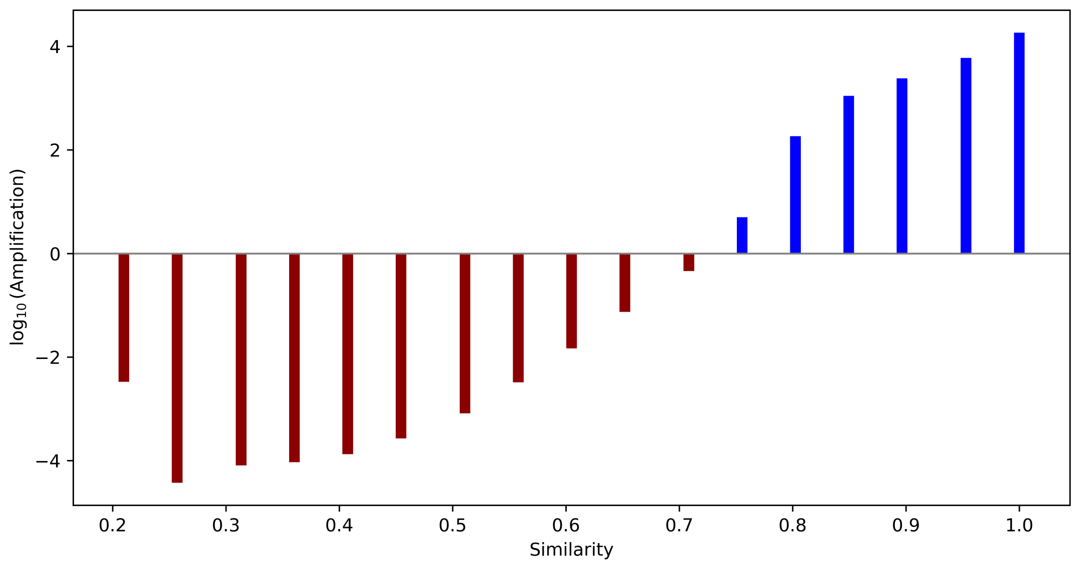


Figure 4.3: Simulation results for NVQWOA applied to seed 2025. Logarithmic scale solution amplification as a function of solution similarity.

This pattern of amplification is confirmed in Figure 4.3. Solutions with similarity below 0.75 are attenuated by the NVQWOA, and solutions with similarity greater than 0.75 are amplified, with the average amplification increasing exponentially with similarity. The similarity scores of the solutions with the most average attenuation is 0.2593, with an amplification factor of 3.73×10^{-5}. Interestingly, the solutions with the most attenuation are not those with the minimum similarity scores. The two minimum quality solutions, which have a similarity of 0.21, received an average amplification of 3.33×10^{-3}, which is more amplification than received by any solution with similarity score less than 0.6. This reduction in attenuation is due to the effect of applying the phase-separation unitary U_Q. The lowest quality solutions receive the smallest phase shifts from U_Q, while the highest quality solutions receive the largest phase shifts from U_Q. This implies that out of all the solutions which receive attenuation, the lowest quality solutions have phases closest to the optimal solutions. Since phase encodes the quality of solutions, the lowest quality solutions are encoded similarly to solutions with higher quality, which increases the amount of amplification they receive from the NVQWOA.

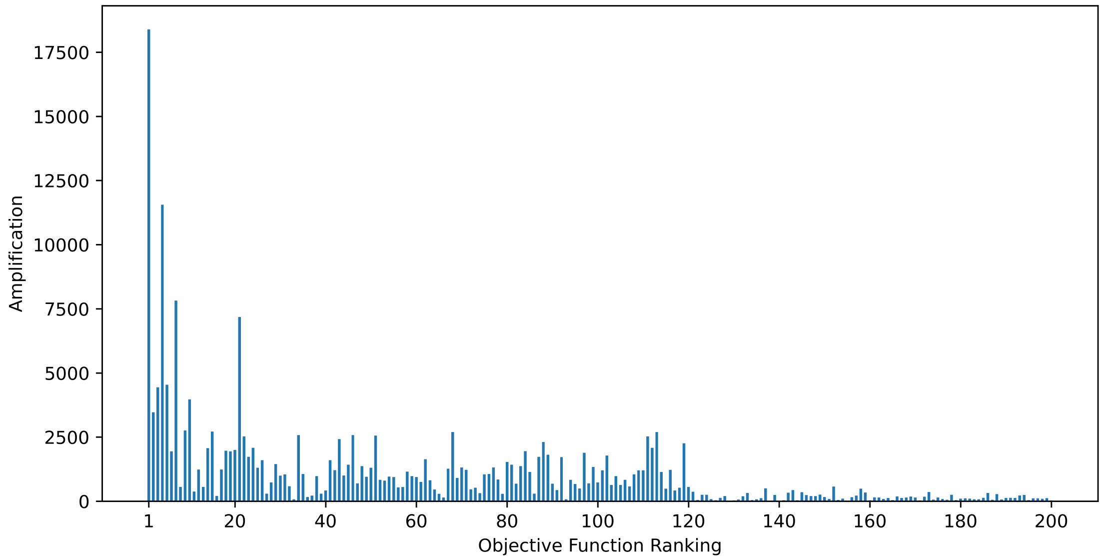


Figure 4.4: Simulation results for NVQWOA applied to seed 2025. Amplification applied to the best 200 solutions.

Analysing the pattern of amplifications near the best solutions, Figure 4.4 verifies that no individual solution receives greater amplification than the optimal solution, as expected, with an amplification factor of 18391. The solution with the second-most amplification (factor of 11552) is ranked 4 in the figure, but the solutions ranked 2 to 5 all had identical similarity scores of 0.95, so its amplification relative to the others is just caused by sorting degeneracy. Solutions ranked 6 to 21 had similarity scores of 0.90, and these were the last solutions to receive amplification greater than 5000.

The standard deviation of the amplifications for the top similarity scores in descending order was 0, 3234, 2153, and 663. The standard deviation of 0 corresponds with the fact that there was only one optimal solution. The standard deviations continue to decrease with reducing average amplitude such that the ratio of standard deviation to average amplification is constantly ≈ 1.

4.2 Multiple Graph Analysis

The next phase in the analysis was to perform simulations on many different graphs and observe how the algorithm performs in the average case. For these simulations, the number of iterations was decreased to $p = 10$, and numerical optimisation was used to find suitable hyperparameters for each problem. When referring to the hyperparameter optimisation objective functions explained in section 3.5, we use the acronyms EV for expectation value, OSP for optimal solution probability, and CVaR for conditional value at risk.

4.2.1 Algorithm Behaviour and Amplification

Expanding on the findings from the previous section, probability distributions for the similarities were analysed. Figures 4.5 and 4.6 show the averaged probability distributions over the possible similarity scores, using a linear probability scale for Figure 4.5 and a log probability scale for 4.6. Figure 4.7 shows the mean amplification applied to solutions at each similarity.

After amplification, the probability distributions (4.5 and 4.6) are skewed towards higher-similarity solutions compared to the initial distribution. The EV distributions correlate strongly with the single graph example from the previous section, but their modes shifted even further, moving from an initial mode of 0.57, 0.56, 0.57 for $n = 8, 9, 10$ respectively, to 0.88, 0.91, 0.85 after amplification. The initial modes closely match the example from the previous section (0.56), but the final modes are greater than the final mode from there (0.80). This greater shift in the mode of the distribution, despite using far fewer iterations, is most likely due to the optimised hyperparameters enhancing the amount of amplification applied to optimal and near-optimal solutions.

As expected, the probability to measure the optimal solution is increased the most by the OSP distributions for each value of n, but unexpectedly, this does not appear to skew the probability distribution as much as EV or CVaR. Instead, there is a sudden increase in probability at the optimal solution, but the rest of the distribution is only slightly moved, with the main distribution modes at 0.69, 0.71, 0.69. The OSP optimisation can only observe the probability to measure an optimal solution, so any probability distributed from low-quality solutions must pass through the near-optimal solutions and greatly affect the optimal solution, and the probability from those near-optimal solutions will also be channelled into the optimal solution. This reduces the amount of probability the near-optimal solutions have in the final distribution.

The CVaR distributions also possess distinctive features. For $n = 8$, the mode is at the optimal solution and the distribution is easy to interpret as applying increasing amounts of amplification to higher similarity distributions, but for $n = 9$ and $n = 10$, the modes are at 0.91 and 0.93 respectively and these modes have much higher probability than anywhere else on the otherwise smooth and well-behaved distributions. These features are caused by the averaging over the 30 different problem instances at each n. In our analysis, we found that when individual problem instances are analysed, the hyperparameters that CVaR converges

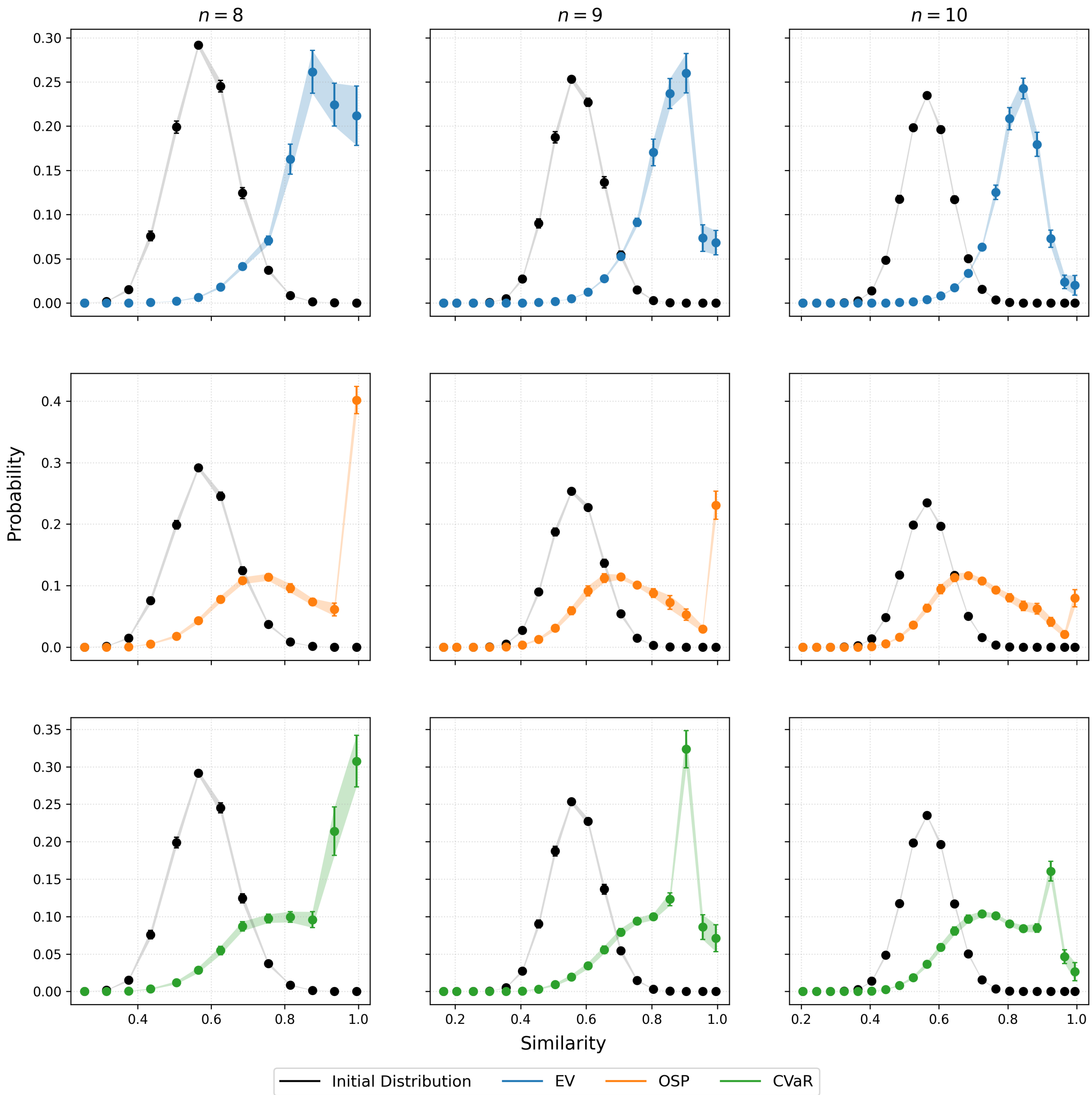


Figure 4.5: Linear-scale probability distributions for similarity as measured from the initial equal superposition state (black) and the final amplified state. Average simulation results for NVQWOA applied to 30 graphs at (left) $n = 8$, (center) $n = 9$, (right) $n = 10$. (Top) hyperparameters optimised for EV. (Middle) hyperparameters optimised for OSP. (Bottom) hyperparameters optimised for CVaR.

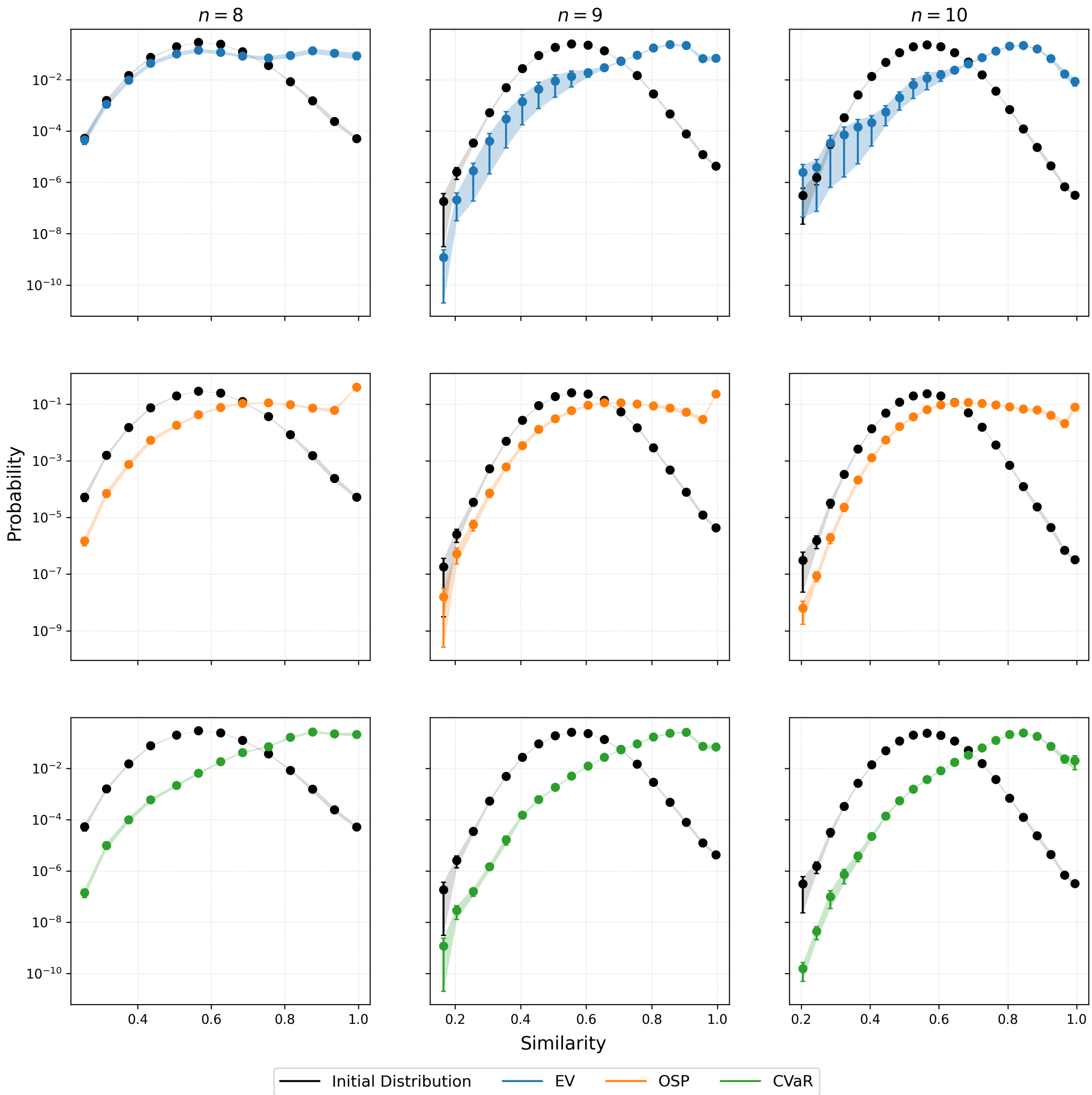


Figure 4.6: Log-scale probability distributions for similarity as measured from the initial equal superposition state (black) and the final amplified state. Average simulation results for NVQWOA applied to 30 graphs at (left) $n = 8$, (center) $n = 9$, (right) $n = 10$. (Top) hyperparameters optimised for EV. (Middle) hyperparameters optimised for OSP. (Bottom) hyperparameters optimised for CVaR.

to are often near the hyperparameters that EV converges to or the ones that OSP converges to. Usually CVaR does not converge to its own set of clearly distinct values. This causes the probability distributions that CVaR creates on individual problem instances to closely match with the probability instances of EV or OSP. Therefore, CVaR's averaged probability distributions in Figures 4.5 and 4.6 display a mix of the main features present in the EV and OSP graphs.

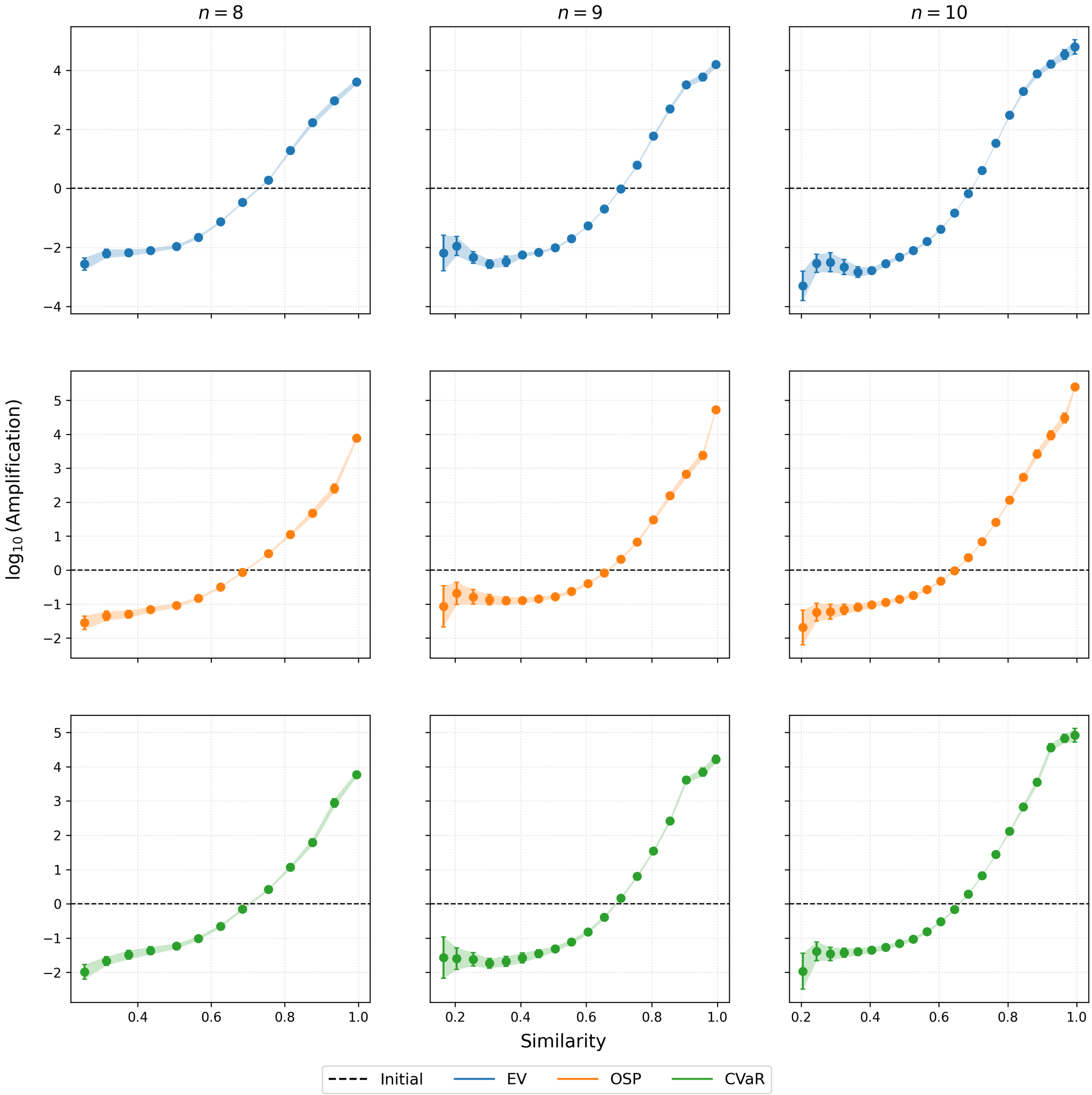


Figure 4.7: Logarithmic scale solution amplification as a function of solution similarity. Average simulation results for NVQWOA applied to 30 graphs at (left) $n = 8$, (center) $n = 9$, (right) $n = 10$. (Top) hyperparameters optimised for EV. (Middle) hyperparameters optimised for OSP. (Bottom) hyperparameters optimised for CVaR.

The average amplification distributions, shown in Figure 4.7, expands on the analogous plot from the previous section (Figure 4.3). A point of discussion about that plot was that the solutions with the lowest qualities are expected to receive less attenuation than other low-quality solutions. In Figure 4.7, however, the solutions with with lowest appear to receive *more* attenuation than other low-quality solutions. This is caused by the attainable values of similarity differing between problem instances. Over several problem instances, the range of possible values for the similarity may be narrower or wider due to the structure of the particular problem graph (For example, a complete graph has only one possible value for its edge overlap over all possible vertex mappings). Graphs with a narrower range of possible similarities affect the average similarity within their subshell on the mixing graph. The degeneracy between certain permutations causes solutions that are a greater away from any reference solution on the mixing graph to not necessarily have a different objective function value. This reduces the adherence to the second condition of the NVQWOA, as discussed in section 2.6.2.2, which means the NVQWOA does not perform as well. The greater the objective function degeneracy in the solution space, the worse the algorithm will perform. Therefore, the algorithm performs best on the problem instances with the fewest degenerate permutations, which corresponds to the problem instances with the widest range of possible similarity values. This effect appears to increase with problem size, which is expected as larger graphs are more likely to possess degenerate permutations.

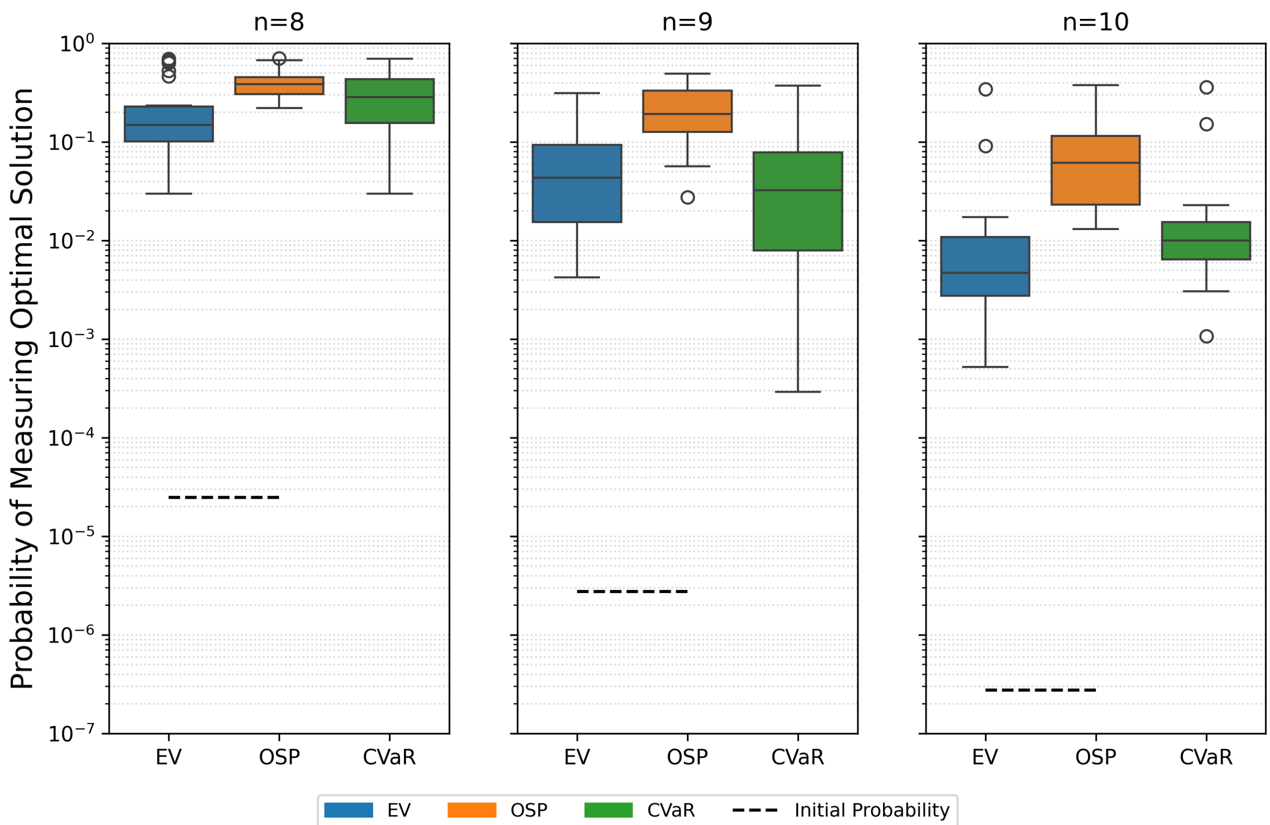


Figure 4.8: Simulation results of the NVQWOA applied to 30 graphs at (left) $n = 8$, (center) $n = 9$, (right) $n = 10$. Boxplots represent the distribution in probability to measure the optimal solution.

It is evident from Figure 4.8 that the NVQWOA performs well in the average case to increase the probability of measuring the optimal solution. In the previous section, the fixed hyperparameter problem amplified its optimal solution probability from 2.76×10^{-6} to 0.051 over $p = 30$

iterations. The optimised hyperparameters in the equal-size $n = 9$ problem set only received $p = 10$ iterations, but they still achieved a final optimal solution probability of 0.068 for EV, 0.230 for OSP, and 0.071 for CVaR. OSP is expected to perform well in this measure, but the values for EV and CVaR provide strong evidence for the role of hyperparameter optimisation in the NVQWOA.

4.2.2 Distance Analysis

We next analysed the distance between each solution and the optimal solution(s) to determine the degree to which amplification favours solutions close to the optimum in the solution space. The measures of distance used were Hamming distance and subshell distance, described in section 3.6.2.

We calculated the mean Hamming distance before and after amplification for each problem instance and for each hyperparameter optimisation objective function. The change in the mean Hamming distance before and after amplification is represented as the boxplots shown in Figure 4.9. On average, the algorithm was successful at significantly reducing the mean Hamming distance. However, the plots show that the variance in these distributions was large, and the Hamming distance was not significantly changed for certain problem instances. The minimum improvement in mean Hamming distance was mostly unaffected by increasing problem size, which may suggest that the source of variance was the structure of the problem instances.

Figure 4.10 shows how much amplification was applied to each solution based on their Hamming distance. The minimum Hamming distance to a non-optimal solution is 2, because the solutions to the graph similarity problem consist of permutations. More concretely, vertex mappings swap the labels of two vertices, so the minimal non-identity permutation differs from the identity permutation in two postions. The optimal solutions when optimising for EV received average amplifications of 4385, 14875 and 39893 for $n = 8, 9, 10$. The $n = 9$ fixed hyperparameter case applied an amplification of 18391 to its optimal solution, so on average, EV hyperparameter optimisation did not apply more amplification to the optimal solution. However, the fixed parameter case was ran for $p = 30$ iterations and actually had an amplification of 6819 when $p = 10$. Deeper quantum circuits are desirable, but these results show we can achieve nearly the same results with shallower circuits that implement hyperparameter optimisation. For instance, CVaR and OSP both outperformed EV, applying more amplification to the optimal solution and more attenuation to distant solutions.

We also analysed the subshell distance between each solution and the optimal solution(s), which directly reflects the structure of the permutation mixer and the connectivity of the mixing graph. Figure 4.11 is similar to Figure 4.10, but plots amplification against subshell distance instead of Hamming distance. Comparing the two plots, we see that the amplification is monotonically decreasing with subshell distance, but is not monotonically decreasing with Hamming distance and is generally less smooth. This is because the subshell distance is more naturally related to

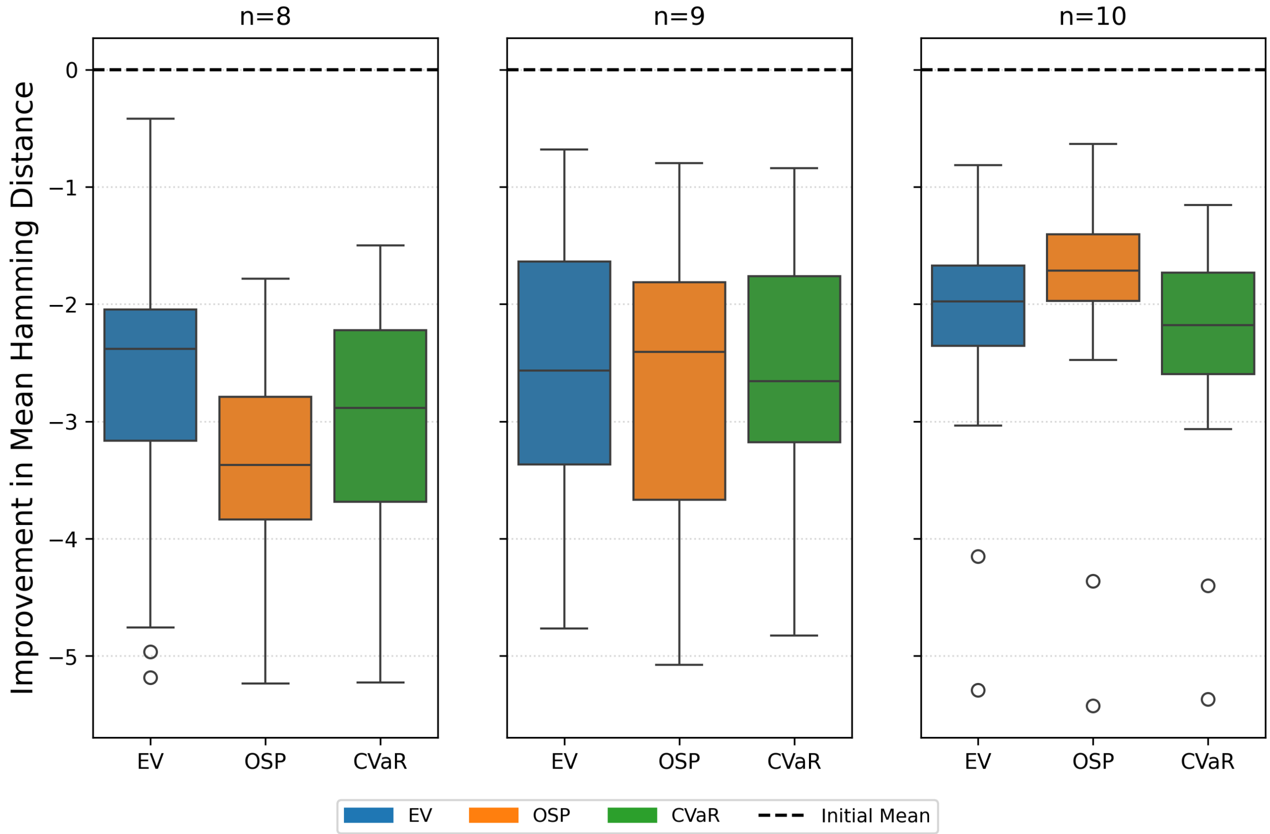


Figure 4.9: Simulation results of the NVQWOA applied to 30 graphs at (left)$n = 8$, (center) $n = 9$, (right) $n = 10$. Boxplots representing the distribution of improvements in mean Hamming distance. The initial mean (black) represents the mean Hamming distance before amplification was applied.

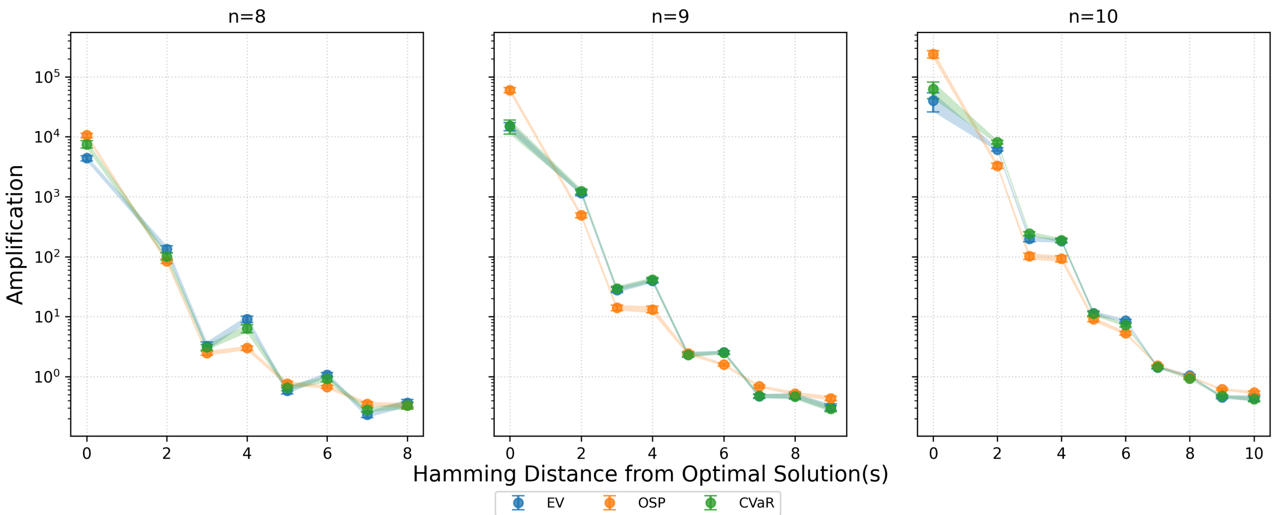


Figure 4.10: Mean amplification applied to each solution grouped by their Hamming distance from the optimal solution.

the structure of the mixer. The Hamming distance can give the same value for solutions that are a different distance away on the mixing graph. For example, $(1, 2, 3, 4)$ has a Hamming distance of 4 from both $(2, 1, 4, 3)$ and $(2, 3, 4, 1)$, but the subshell distances would be 2 and 3 respectively. We can see this degeneracy in Figure 4.10 starting at a Hamming distance of 4. In contrast, the subshell distance directly encodes the distance between solutions on the mixing graph, and the distance determines how much complex amplitude a solution receives from other solutions.

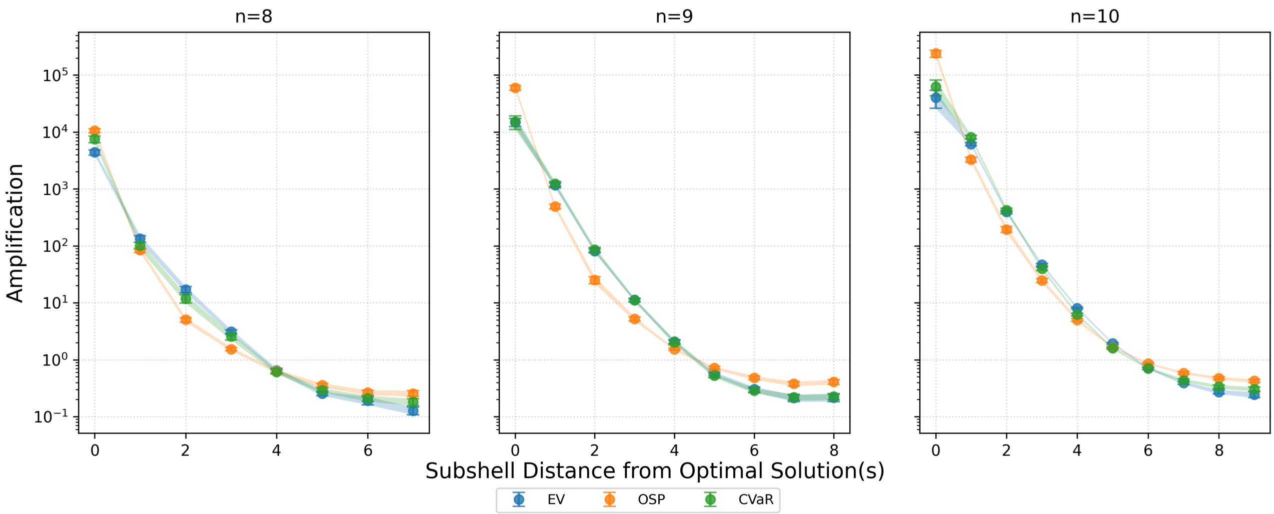


Figure 4.11: Mean amplification applied to each solution grouped by their subshell distance from the optimal solution.

4.2.3 Hyperparameter Analysis

As described in section 3.5, we optimised the three NVQWOA hyperparameters β, γ, t under three different objective functions applied to the final state: the expected value (EV), the probability of measuring an optimal solution (OSP), and the conditional value at risk (CVaR). The hyperparameters were initialised as $\gamma = 1.0, t = 0.1, \beta = 0.4$. The mean hyperparameters for each problem size and hyperparameter optimisation objective function are recorded in Tables 4.1, 4.2, 4.3.

Across all problem sizes studied, optimisation according to OSP consistently produced the strongest amplification of the optimal solution, as expected given that it directly targets this quantity. However, this approach is not physically realisable without prior knowledge of the solution and is therefore best understood as an upper bound on achievable performance. CVaR-optimised parameters produced results close to this upper bound, typically outperforming EV optimisation in both amplification strength and probability mass concentration around near-optimal solutions. This aligns with previous findings that CVaR focuses the optimisation landscape on the most important region of the state space, reducing the impact of low-quality solutions and accelerating convergence [41, 42]. EV optimisation, while yielding the lowest amplification overall, remains the most experimentally accessible because it relies only on measurements of the expectation value and requires significantly fewer samples. Fixed hyperparameters

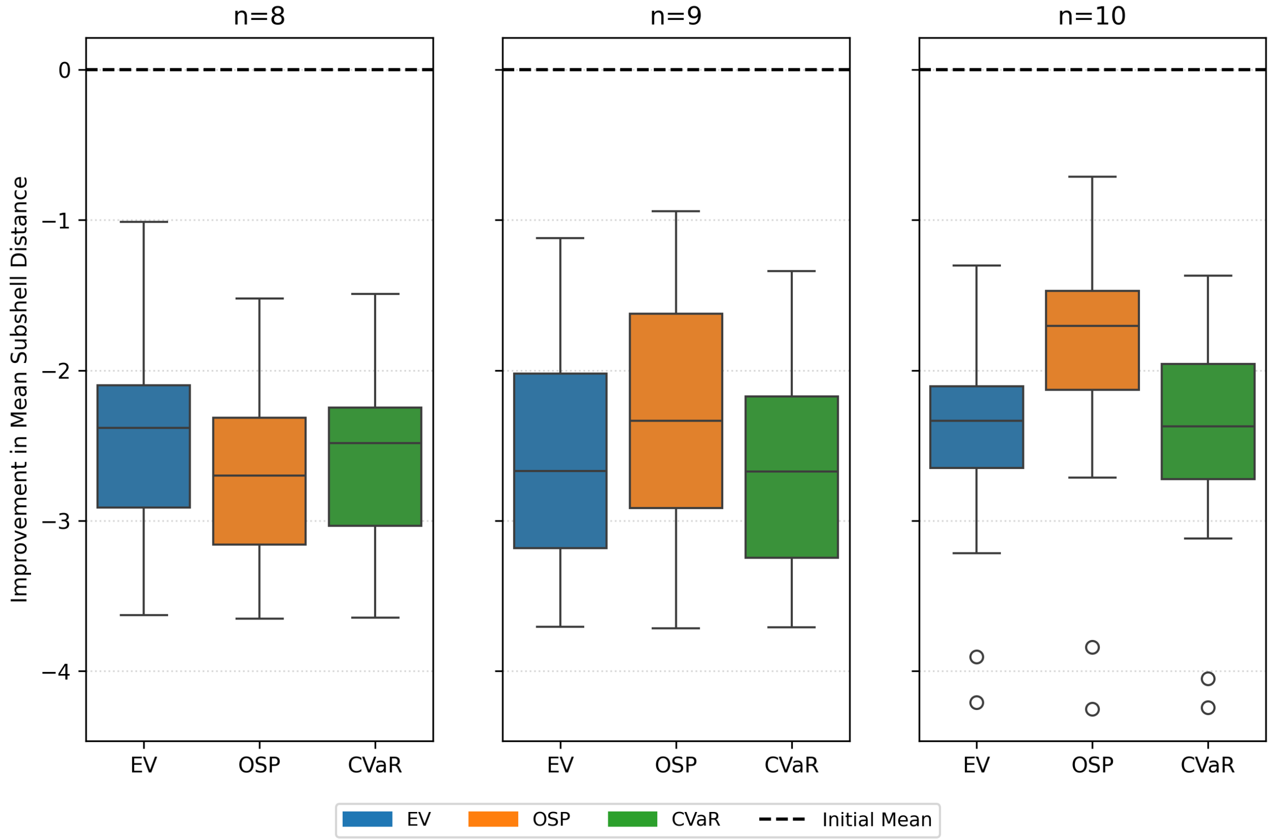


Figure 4.12: Improvement in subshell distance boxplot.

were consistently outperformed by optimised hyperparameters, even though the fixed case was a much deeper circuit at $p = 30$ iterations while the optimised set had only $p = 10$.

The optimised hyperparameter values exhibited relatively low variance across problem instances, suggesting a degree of robustness in the algorithm's hyperparameter landscape. In particular, the values of γ and t suggest a scaling trend with increasing problem size, γ increasing and t decreasing. This behaviour likely reflects how as the solution space increases in size, the applied phase separation needs to increase to apply more amplification, and the mixing times need to decrease to minimise the probability amplitude distributed from the optimal parameters to the suppressed sub-optimal solutions. The parameter β, which determines how the magnitudes of γ and t change during the algorithm, remain comparatively stable across all instances, suggesting that its optimal value is less sensitive to problem size than those of γ and t.

These observations highlight two key points. First, although precise tuning can enhance performance, the NVQWOA's amplification is not sharply peaked around a single set of hyperparameters, implying a broad region of parameter space where useful amplification occurs. This property reduces the classical overhead associated with hyperparameter tuning, which is a major limitation of variational approaches such as the QAOA. Second, the differing performance characteristics of the three hyperparameter optimisation objective functions illustrate a trade-off between performance and feasibility. EV optimisation offers a realistic path to implementing the algorithm on near-term quantum hardware, while CVaR provides some benefits to performance if more advanced measurement and feedback strategies are deployed, and OSP serves as an unfeasible upper bound for the best-case hyperparameters.

$n = 8$	β	γ	t
EV	0.55(2)	1.15(3)	0.169(3)
OSP	0.78(2)	1.01(2)	0.180(2)
CVaR	0.71(2)	1.05(3)	0.173(3)

Table 4.1: Hyperparameter optimisation results for $n = 8$.

$n = 9$	β	γ	t
EV	0.586(8)	1.19(2)	0.142(2)
OSP	0.81(2)	1.05(2)	0.164(2)
CVaR	0.73(2)	1.16(2)	0.147(2)

Table 4.2: Hyperparameter optimisation results for $n = 9$.

$n = 10$	β	γ	t
EV	0.580(6)	1.28(1)	0.124(2)
OSP	0.78(2)	1.12(2)	0.151(1)
CVaR	0.816(9)	1.13(1)	0.135(1)

Table 4.3: Hyperparameter optimisation results for $n = 10$.

4.3 Subshell Analysis

The final analyses investigated the underlying graph similarity problem structure and the suitability of the permutation mixer for these problems. The calculations were performed using the same set of problems as the previous section, 30 problems for each of $n = 8, 9, 10$.

The suitability of the permutation mixer for the unlabelled graph similarity problem is supported by the quality gap and variance analyses. Figure 4.13 shows the mean quality gap between the optimal solutions and the subshells. For all problem instances, the mean quality gap was a monotonically increasing function with subshell distance, adhering to the second condition of the mixer, that solutions further from the optimum are of systemically lower quality than solutions close to the optimum. This structure enables the NVQWOA to exploit the mixing process effectively, as described in section 2.6.2.2.

The phase-coherence of the structure within each subshell is supported by Figure 4.14, which shows the amount of variance in the objective values for each subshell. Lower variance indicates the solutions within the shell are of similar quality to one another. The optimal solutions always have a similarity of 1 in the problems we constructed, so the variance within subshell $h = 0$ is always 0. The maximum subshell variance was 0.0127, which was in subshell $h = 1$, problem

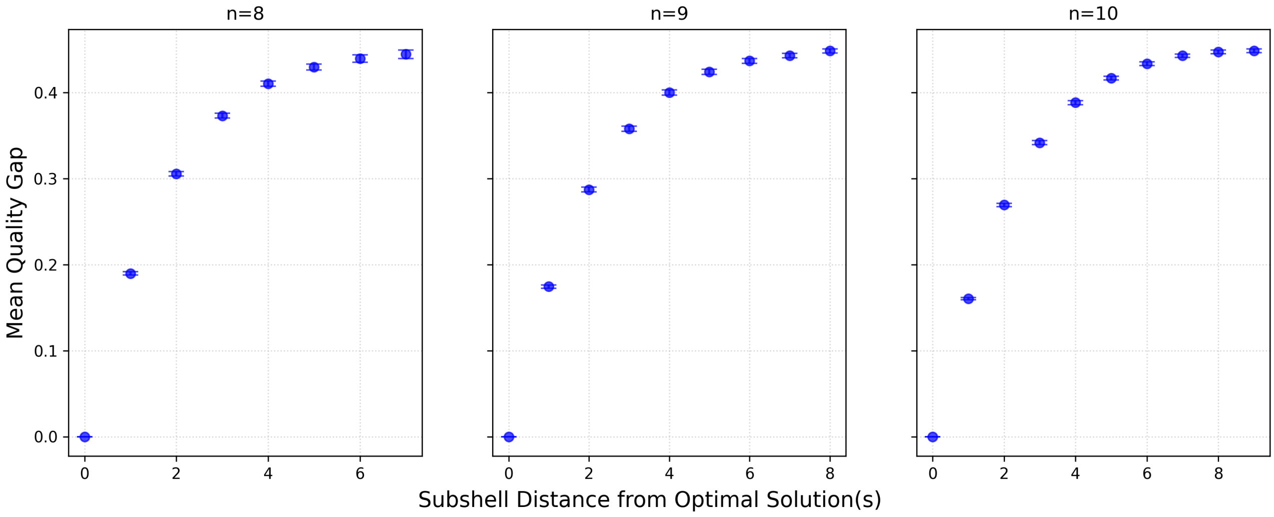


Figure 4.13: Mean quality gap vs subshell distance

seed = 20. This suggests that each subshell should remain phase coherent and adhere to the second condition of the NVQWOA.

The second-order variances were calculated for $n = 8, 9, 10$ to be $4.7\times10^{-6}, 3.2\times10^{-6}, 2.0\times10^{-6}$. These small second-order variances imply similar levels of variance across the subshells. The spread of qualities will not significantly change between subshells, just the mean. Pairing this with a monotonically increasing mean cost gap over subshell distance implies each subshell behaves similarly, so no matter where the optimal solution is in the mixing graph, a good choice of hyperparameters will successfully amplify the optimal solution.

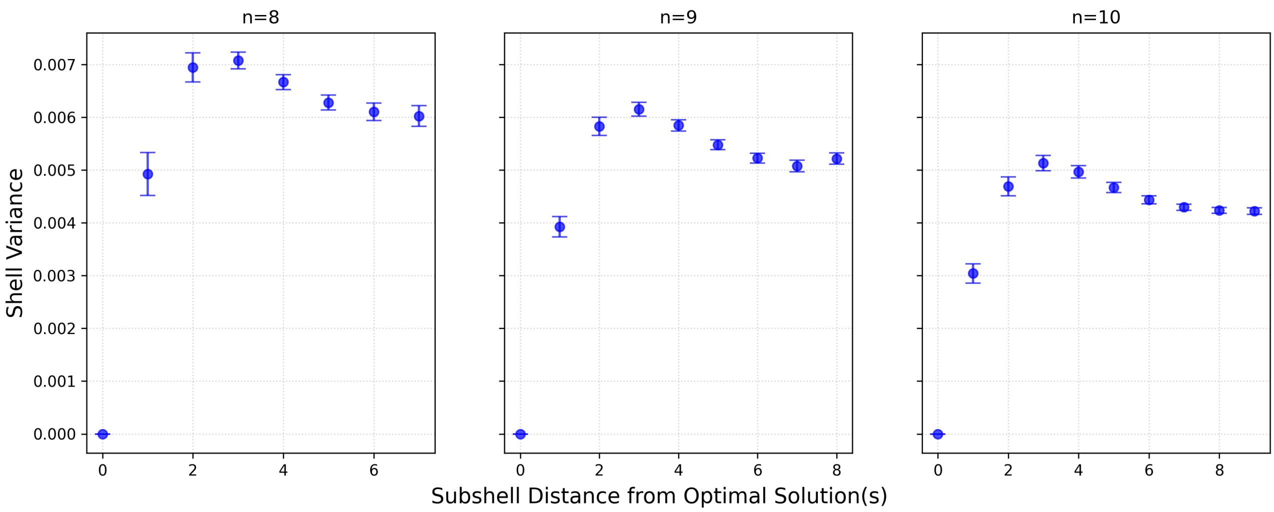


Figure 4.14: Shell variance vs subshell distance

Chapter 5

Future Work

The results of this thesis suggest several promising directions for future research.

Algorithmic Extension

One natural next step is to explore alternative graph similarity measures beyond the edge overlap measure used here. Measures such as graph edit distance or maximum common subgraph could provide a more nuanced assessment of graph similarity and may yield different amplification behaviours under NVQWOA dynamics. Investigating how the algorithm performs with cost functions that include structural penalties or weightings could also reveal how sensitive it is to the choice of objective and inform the design of mixers tailored to specific graph properties.

Methodological Advances

The present study was limited to small problem sizes by the computational cost of state-vector simulation. Extending the analysis to larger graphs using tensor-network simulation or distributed simulation frameworks would provide insight into the algorithm's scaling behaviour. Additionally, implementing NVQWOA on noisy intermediate-scale quantum (NISQ) hardware would allow experimental validation of its performance under realistic noise and decoherence conditions. Such tests could help assess the robustness of amplification under hyperparameter selections and inform strategies for error mitigation or noise-aware parameter selection.

Incorporating Problem Structure into the Initial State

In the present formulation, NVQWOA begins from an initial superposition over all possible solutions, which treats all states as equally likely. Although this is easy to implement, it gives up the opportunity to exploit graph structures that can be easily read from any vertex mapping, such as degree distribution or spectral properties. Using these to inform the structure of the initial state could strongly constrain the solution space of plausible vertex mappings, which would reduce the search space that the algorithm explores.

Chapter 6

Conclusions

This thesis has investigated the application of the non-variational quantum walk-based optimisation algorithm (NVQWOA) to the unlabelled graph similarity problem, a fundamental challenge in combinatorial optimisation with broad relevance to network science, logistics, and bioinformatics. Unlike variational quantum algorithms such as QAOA, which rely on the repeated optimisation of large parameter spaces, NVQWOA operates with a fixed set of three hyperparameters and leverages continuous-time quantum walk dynamics to redistribute probability amplitude in a way that favours high-quality solutions. Through classical state-vector simulations, this work has characterised the performance of NVQWOA, examined how its dynamics reflect the structure of the solution space, and evaluated how its behaviour depends on hyperparameter choice and objective function.

The results presented here demonstrate that NVQWOA successfully amplifies the probability of optimal and near-optimal vertex mappings compared to the uniform initial state. Amplification is not confined to a single solution but extends across a neighbourhood of structurally related configurations, a property that is especially valuable when near-optimal solutions are of practical interest. Analysis of solution distance metrics shows that amplification strength correlates strongly with proximity to the optimum, and that subshell distance, which reflects adjacency in the permutation graph, captures this structure more effectively than simple Hamming distance. Investigations of mean quality gaps and variance trends further reveal that the solution space is organised in a way that satisfies the theoretical conditions necessary for efficient optimisation.

The study of hyperparameter behaviour shows that while NVQWOA's performance can be enhanced through optimisation, it is not narrowly dependent on precise parameter values. Optimal γ and t values exhibited signs of trends with problem size, while β remains largely stable, indicating a relatively smooth parameter landscape. This robustness, combined with the algorithm's small number of tunable parameters, significantly reduces classical optimisation overhead compared to variational approaches. Among the objective functions tested, conditional value at risk (CVaR) emerged as a particularly effective choice, striking a balance between solution amplification and experimental feasibility, while expected value (EV) optimisation offers a straightforward path for implementation on near-term hardware.

These findings establish NVQWOA as a promising approach to solving the unlabelled graph similarity problem. Its sensitivity to solution structure, exploitation of structure using problem-specific mixers, and minimal parameterisation make it well suited for execution on noisy intermediate-scale quantum (NISQ) devices, where deep circuits and extensive classical feedback are impractical.

This thesis contributes both a detailed characterisation of NVQWOA's behaviour and a demonstration of its potential as a non-variational alternative to existing quantum optimisation approaches. By showing that NVQWOA can exploit problem structure to focus computational effort on high-quality regions of the solution space with minimal classical overhead, this work lays the foundation for future research into scalable, structure-aware quantum optimisation algorithms and their application to graph similarity and beyond.

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Appendix A

Research Proposal

Application of a Non-variational Quantum Walk-based Optimisation Algorithm for Graph similarity

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May 26, 2024

A.1 Introduction

A.1.1 Background

Complex systems characterised by extensive interconnectivity, such as social networks, electrical power grids, circulatory systems, and database architectures are frequently modelled using network graphs [23]. Measuring the similarity between these graphs is crucial for enabling quantitative comparisons and gaining valuable insights. However, challenges arise when dealing with very large graphs, which significantly impact the scalability and efficiency of existing similarity algorithms. Moreover, the graph similarity problem becomes computationally intractable (requiring exponential time) for graphs without known labelling using conventional methods [3]. This project aims to address these challenges by exploring the application of the non-variational quantum walk-based optimisation algorithm (non-variational QWOA) [16], potentially offering a breakthrough in solving the graph similarity problem more efficiently and effectively.

A.1.2 Literature Review

Network graph models have become crucial components in systems that are used in everyday life, such as search engines and social networks [22]. Graph similarity (or graph distinguishability, as it is sometimes referred [21]) is often important in these contexts. For example, social networks are compared to identify certain interaction patterns [44], traffic networks are compared to help detect abnormal changes in traffic patterns [5], and web networks are compared for anomaly detection [4]. In all these cases, the similarity between two graphs with overlapping sets of nodes is assessed and the detection of changes in the patterns of connectivity is important. For graphs which match approximately, it is useful to obtain a quantitative measure of similarity. For example, classification techniques in machine learning match objects that are within a certain tolerance.

Graph similarity measures are quantitative calculations based on comparisons made between the structure of network graphs. Different measures of graph similarity will produce a variety of results due to differences in how the structures of the graphs are analysed. There are many measures of graph similarity, including Maximum Common Subgraph [45], Graph Edit Distance [8], Frobenius Distance [46], and Graph Eigendecomposition [47]. These measures are all successful at indicating the degree to which two graphs are similar and can also detect small differences between them.

Graph similarity in detail:

Let $G_1(V, E_1), G_2(V, E_2)$ be two graphs defined by a set of vertices V and edge sets E_1, E_2. Define $sim(G_1, G_2) \in [0, 1]$ as the similarity of these graphs and define A_1, A_2 as the adjacency matrices of G_1 and G_2, respectively.

Koutra et al. formalised a set of axioms that graph similarity measures should conform to [24]:

1. Identity: $sim(G_1, G_2) = 1 \iff G_1 \cong G_2$
2. Symmetric: $sim(G_1, G_2) = sim(G_2, G_1)$
3. Zero: For a clique graph G_1 and a empty graph G_2, $sim(G_1, G_2) \to 0$ as $n \to \infty$

They also raise that similarity algorithms should factor in the structural importance of each edge, the weight of removed edges, and the total number of edges in a graph. They also say a good graph similarity measure should be able to distinguish targeted and random changes, and should be scalable to large, generated graphs. However, this scalable criterion has only been fulfilled by algorithms applied to graphs with known node correspondence.

The failure of the scalability criterion for algorithms applied to graphs with unknown node correspondence (hereafter referred to as unlabelled graphs) is due to problem of finding the correct corresponding vertex labels between the graphs. Observe that due to Axiom 1, similarity algorithms should be able to identify isomorphic graphs (and assign them a similarity of 1), but confirming isomorphism requires checking each permutation of vertex labelling. Using this fact, the similarity of an unlabelled G_1 and G_2 can be expressed as:

$$sim(G_1, G_2) = \max_{\pi} simscore(A_1^{\pi}, A_2)$$

Here, π ranges over all permutations of the vertex set V, A_1^{π} denotes the adjacency matrix obtained from G_1 by permuting rows and columns according to π, and *simscore* is a function that assigns a scalar in the range $[0, 1]$, called the similarity score, to a particular pair of adjacency matrices. The structure of *simscore* depends on the measure of graph similarity that is being used. Note that in this scope, the graph similarity problem is the problem of finding the optimal alignment of two graphs such that their similarity score is maximised.

Since the unlabelled graph matching problem is NP-complete [3], existing algorithms to measure graph similarity take exponential time to compute. Therefore, it is desirable to use quantum algorithms.

Quantum Optimisation Algorithms:

Using the previously established framework, we can reduce the problem of graph similarity to a combinatorial optimisation problem (COP) suitable for quantum algorithms. The COP consists of n integer decision variables $x_i \in \{0, 1, 2, ..., n-1\}$ where the solutions are characterised by vectors $\mathbf{x} = (x_1, x_2, ..., x_n)$, which represent a particular labelling of vertices of G_1 onto G_2 (similar to A_1^{π}), and the space of feasible solutions S contains all such vectors without repeating elements, such that there are N feasible solutions where $N = n!$. The system is initially in an equal superposition of all solutions,

$$|s\rangle = \frac{1}{\sqrt{N}} \sum |\mathbf{x}\rangle \,,$$

and we seek to amplify the probability of measuring the solution $\mathbf{x}$ that maximises $f(\mathbf{x})$, where f is the objective function that represents the equivalent *simscore* for a particular $\mathbf{x}$.

One potential quantum algorithm that could be used is the quantum approximate optimisation algorithm (QAOA) [12], which uses a variational approach to solving COPs. The algorithm repeatedly applies two Hamiltonians to $|s\rangle$, where the application times of each Hamiltonian are classically controlled parameters. These parameters are initially random and must be tuned via a classical optimisation process. The parameters that globally maximise the expected value of the objective function must be selected for optimal performance of the algorithm [13].

Another variational algorithm is the quantum walk-based optimisation algorithm (QWOA), which is closely related to the main algorithm of this project. The QWOA is a more general form of the QAOA, where the time-based application of the transverse-field Hamiltonian is replaced by a continuous-time quantum walk (CTQW) on a graph that contains basis states encoding the feasible solutions to the COP [48]. The other Hamiltonian is unchanged and phase-shifts basis states depending on their associated objective function values.

The first concern in the use of variational algorithms is that the process of tuning the parameters is costly. The requirement to globally maximise these parameters means that gradient-based methods are more sensitive to the initial parameters, but the space of possible initial parameters grows exponentially in parameter count [14]. In essence, variational algorithms solve the maximisation of the objective function, but also present the challenge of solving the maximisation problem of their own parameters. As such, a non-variational approach is preferred for our problem.

Another problem with the above algorithms is that they are built on a quadratic unconstrained binary optimisation (QUBO) framework, and as such, the decision variables x_i must be binary. This conflicts with the structure of the graph similarity problem, which requires the use of non-binary decision variables. Although studies have shown that non-binary problems can be implemented into the QUBO framework [18], it requires introducing a penalty component to the objective function, further increasing the number of parameters. As such, an approach not restricted to only binary decision variables is preferred.

To address these issues, this project will implement a non-variational, non-binary-restricted quantum algorithm.

The Non-variational Quantum Walk-based Optimisation Algorithm:

The main algorithm of this project is the non-variational quantum walk-based optimisation algorithm (non-variational QWOA) [16], which replaces the QWOA CTQW graph with a mixing graph that connects nearest-neighbour solutions.

For the graph similarity problem, the mixing graph connects each solution state to the set of $\frac{1}{2}n(n-1)$ possible transpositions (obtained by swapping two elements in $\mathbf{x}$). As such, the adjacency matrix of the mixing graph can be expressed as

$$A = \sum_{i=0}^{n-2} \sum_{i+1}^{n-1} \mathrm{SWAP}_{i,j},$$

where $\mathrm{SWAP}_{i,j}$ is defined as the permutation matrix associated with swapping the i^{th} and j^{th} entry in $\mathbf{x}$ and applying identity to the remaining entries.

There are two important unitaries driving the interference process in the non-variational QWOA. The first is the phase-separation unitary,

$$U_Q = e^{-i\gamma Q}.$$

Q is the quality, a diagonal operator such that $Q\left|\mathbf{x}\right\rangle = f(x)\left|\mathbf{x}\right\rangle$. Hence, the phase-separation unitary applies a phase-shift to each solution state proportional to its objective function value.

The second unitary is called the mixing unitary or mixer, and it performs the CTQW for time

t on the mixing graph,

$$U_M = e^{itA}.$$

After establishing these unitaries and given a pre-selected number of iterations p, the amplified state is defined as:

$$|\gamma, t, \beta\rangle = \left[\prod_{j=0}^{p-1} U_M(t_j) U_Q\left(\frac{\gamma_j}{\sigma}\right)\right] |s\rangle$$

Where γ, t, and β are classically controlled positive-valued parameters that determine the applied phase separations and mixing times for each iteration. The values of γ_j increase linearly over the domain $[\beta\gamma, \gamma]$, and the values of t_j decrease linearly over the domain $[\beta t, t]$. σ is standard deviation of $f(x)$, controlling the different possible measures of graph similarity such that $\gamma \approx 1$. An approximate value of σ is sufficient and so can be found through random sampling of $f(\mathbf{x})$. An appropriate value of β, which is in the interval $(0, 1)$, is unique to a particular problem type. As such, it is expected that once found, a single value of β will be suitable for all graph similarity problems. Notably, Bennett et al. observes that optimal values of β range from approximately 0.4 down to $\frac{1}{p}$ for the problems of Weighted Maxcut, Maximum Independent Set, K-Means Clustering, Capacitated Facility Location, and the Quadratic Assignment Problem [16].

A.2 Project

A.2.1 Aims

The primary goal of this project is to explore the uses of the non-variational QWOA for computing graph similarity, particularly investigating how modifications to the objective function and parameter tuning method influence the performance.

Secondary goals include benchmarking the performance of the algorithm using high performance computing (HPC) resources as well as generalising the implementation of the algorithm to work on weighted graphs, directed graphs and graphs with differing numbers of nodes. Additionally, I may show a proof-of-concept application of the algorithm on a practical case, such as small social network similarity.

A.2.2 Significance

This numerical simulation project is part of a larger project to benchmark the non-variational QWOA across various combinatorial optimisation problems. The non-variational QWOA has shown promising preliminary results, which may indicate that it is a practical quantum algorithm that could outperform equivalent classical or quantum methods. By contributing to the development and benchmarking of the non-variational QWOA, this project aims to advance the field of quantum computing and algorithms, potentially impacting areas such as network security, design, and logistics.

A.3 Research Plan

A.3.1 Methodology and Problems

The code developed to run the numerical simulations on local hardware is in Python, due to the need for high clarity and readability to perform investigations of the performance of the non-variational QWOA. An important aspect of this part of the project is the ability to rapidly prototype the simulation in response to the obtained results. The performance of the code on local hardware is not high priority, as the number of qubits a serial program is able to simulate is highly restricted anyway. The HPC benchmarking simulations will be run on the supercomputer Setonix, at the Pawsey Supercomputing Centre, and the best programming language for these benchmarking simulations will be identified as part of the project.

To analyse the effect of changing the measure of graph similarity, the objective function of the non-variational QWOA will be modified to reflect the required calculation associated with the *simscore* for various measures of graph similarity. Comparisons between the performance of the algorithm using different measures of graph similarity will be used to explore how the algorithm is affected by changes in the measure of graph similarity.

The method by which the γ, t, and β parameters are tuned will also be an area of inquiry. This will involve investigating which tuning method produces the greatest amplification in the optimal solution state, as well as attenuation on the non-optimal solution states.

A potential problem that may arise in benchmarking is that due to the overhead that quantum algorithms require, the benefit of a speed up is normally only achieved for problems with a large size n. Thus, the size of the benchmarking simulations would likely need to have a large n. This may involve excessive time or resources which could cause problems with the size of the scope of the project.

A.3.2 Preliminary Work

This semester I have focused on building an in-depth understanding of the non-variational QWOA and similar quantum algorithms. Additionally, I have worked on obtaining preliminary results, building on the code originally developed by Tavis Bennett.

A graph G_1 was randomly generated using an Erdös-Rényi random graph model [36] with $N = 9$ nodes and probability for edge inclusion $p = 0.5$. The second graph G_2 was a copy of the first graph, but with a single edge removed. The implemented objective function was the number of matching entries in the adjacency matrices of the graphs, divided by the total number of entries in the adjacency matrix. Since only one edge was different between G_1 and G_2, that meant two entries in the adjacency matrices A_1 and A_2 differed, giving an expected similarity of $\frac{79}{81} \approx 0.97$

I tuned the γ, t, and β parameters using two different methods. The first tuning method selected the parameters that would maximise the expected value of the objective function, $\langle f(x)\rangle$. The second tuning method selected the parameters that would maximise the probability to measure the optimal solution state, $P(\text{opt})$. These methods produced the values shown in Table A.1.

Tuning Method	γ	t	β
$\langle f(x)\rangle$	1.4745	0.2194	0.3545
$P(\text{opt})$	1.2378	0.2269	0.6568

Table A.1: Optimal parameter values from different tuning methods

Using these values for the parameters, numerical simulations of the non-variational QWOA was applied to G_1 and G_2 with a number of iterations $p = 10$, and the plots A.1, A.2, and A.3 were created.

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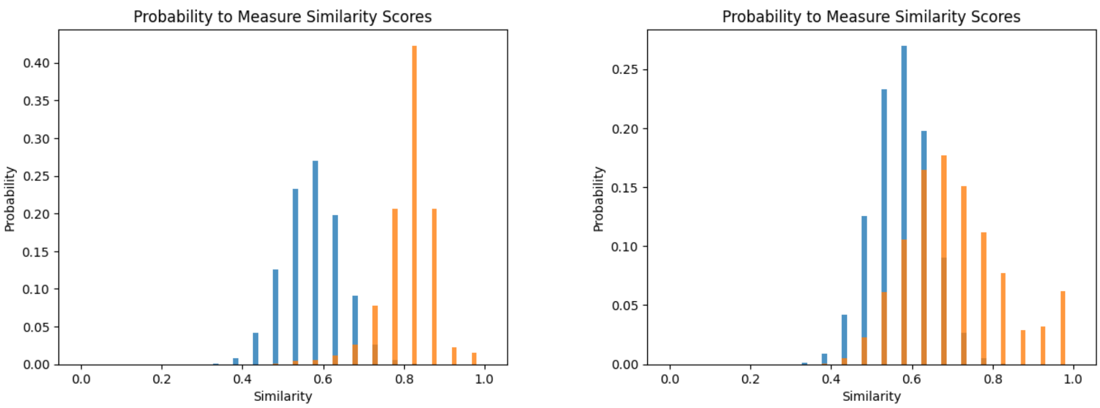

(a) Tuned by $\langle f(x)\rangle$

(b) Tuned by $P(\text{opt})$

Figure A.1: Similarity score probability distributions for different tuning methods.

These plots show that for both tuning methods, the optimal solution was successfully amplified more than any other solution. However, the results also reveal several concerns. Firstly, we expect that tuning for $\langle f(x)\rangle$ and tuning for $P(\text{opt})$ should yield similar parameters, yet they produce very different β values in Table A.1. Bennett et al. suggest that β should be ≈ 0.4 or less, but the value of β obtained from tuning for $P(\text{opt})$ significantly exceeds this guideline.

Secondly, when tuning for $\langle f(x)\rangle$, the amplification of non-optimal solutions is extremely high. The distribution of amplification in A.2 is spread out, and the amount of amplification applied to the optimal solution is much lower than in A.3.

Finally, as a result of these previous behaviours, the most likely similarity score to measure from both simulations is not the similarity score corresponding to the optimal solution.

All of these behaviours warrant further investigation into the configuration of the non-variational QWOA and adaptation of the tuning process is recommended to better align the results with theoretical predictions and practical requirements.

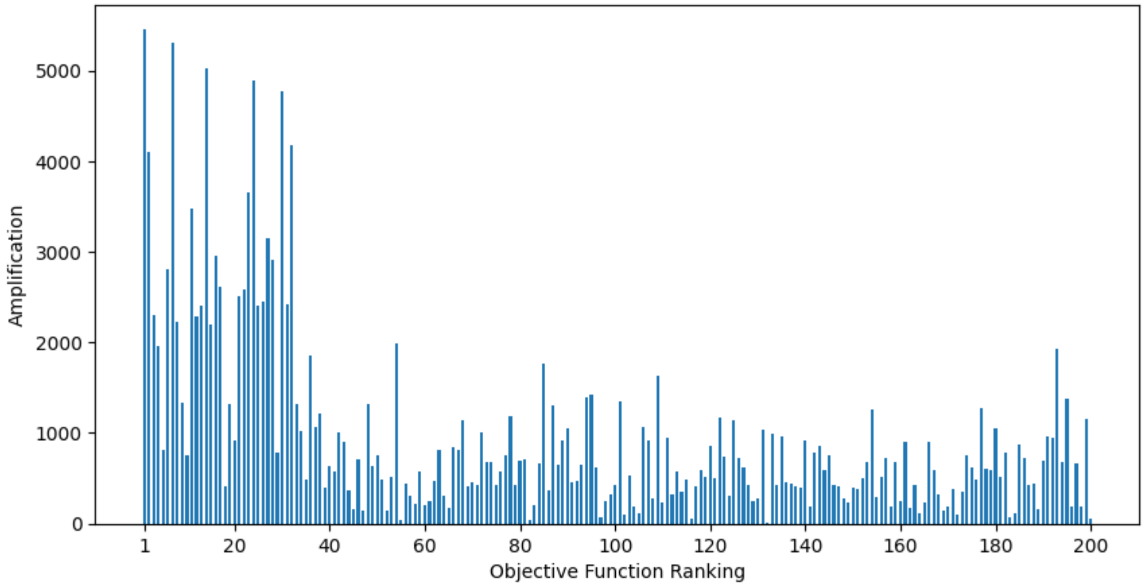

Figure A.2: Amount of amplification applied to the top 200 solutions when tuned by $\langle f(x) \rangle$

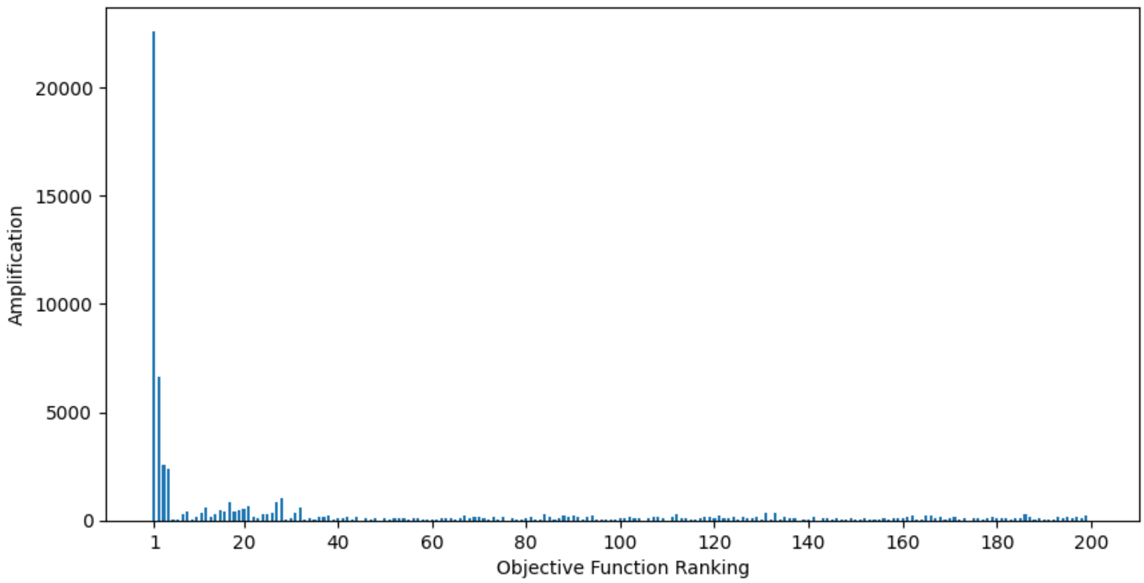

Figure A.3: Amount of amplification applied to the top 200 solutions when tuned by $P(\text{opt})$